SX1508B/SX1509B

World's Lowest Voltage Level Shifting GPIO with LED Driver and Keypad Engine

GENERAL DESCRIPTION

The SX1508B and SX1509B are complete ultra low voltage General Purpose parallel Input/Output (GPIO) expanders ideal for low power handheld battery powered equipment. This family of GPIOs comes in 8-, 16-channel configuration and allows easy serial expansion of I/O through a standard 400kHz I^2C interface. GPIO devices can provide additional control and monitoring when the microcontroller or chipset has insufficient I/O ports, or in systems where serial communication and control from a remote location is advantageous.

These devices can also act as a level shifter to connect a microcontroller running at one voltage level to a component running at a different voltage level, thus eliminating the need for extra level translating circuits. The core is operating as low as 1.425V while the dual I/O banks can operate between 1.2V and 3.6V independent of the core voltage and each other (5.5V tolerant).

The SX1508B and SX1509B feature a fully programmable LED Driver with internal oscillator for enhanced lighting control such as intensity (via 256-step PWM), blinking and breathing (fade in/out) make them highly versatile for a wide range of LED applications.

In addition, keypad applications are also supported with an on-chip scanning engine that enables continuous keypad monitoring up to 64 keys without any additional host interaction reducing bus activity.

The SX1508B and SX1509B have the ability to generate mask-programmable interrupts based on a falling/rising edge of any of its GPIO lines. A dedicated pin (NINT) indicates to a host controller that a state change occurred on one or more of the lines. Each GPIO is programmable via a bank of 8-bit configuration registers that include data, direction, pull-up/pull-down, interrupt mask and interrupt registers. These I/O expanders feature small footprint packages and are rated from -40℃ to +85℃ temperature range.

ORDERING INFORMATION

Part Number	I/Os	Package	Marking
SX1508BIULTRT	8	QFN-UT-20	GAA2
SX1509BIULTRT	16	QFN-UT-28	GBA3
SX1508BEVK	8	Evaluation Kit	-
SX1509BEVK	16	Evaluation Kit	-

KEY PRODUCT FEATURES

- 1.2V to 3.6V Low Operating Voltage with Dual Independent I/O Rails (VCC1, VCC2)
  - Enable Direct Level Shifting Between I/O Banks and Host Controller
- 5.5V Tolerant I/Os, Up to 15mA Output Sink on All I/Os (No Total Sink Current Limit)
- Integrated LED Driver for Enhanced Lighting
  - Intensity Control (256-step PWM)
  - Blink Control (224 On/Off values)
  - Breathing Control (224 Fade In/Out values)
- On-Chip Keypad Scanning Engine
  - Support Up to 8x8 Matrix (64 Keys)
  - Configurable Input Debouncer
- 8/16 Channels of True Bi-directional Style I/O
  - Programmable Pull-up/Pull-down
  - Push/Pull or Open-drain outputs
  - Programmable Polarity
- Open Drain Active Low Interrupt Output (NINT)
  - Bit Maskable
  - Programmable Edge Sensitivity
- Built-in Clock Management (Internal 2MHz Oscillator/External Clock Input, 7 clock values)
  - OSCIO can be Configured as GPO
- 400kHz I^2C Compatible Slave Interface
- 4 User-Selectable I²C Slave Addresses
- Power-On Reset and Reset Input (NRESET)
- Ultra Low Current Consumption: 1uA Typ
- -40℃ to +85℃ Operating Temperature Range
- Up to 2kV HBM ESD Protection
- Small Footprint Packages
- Pb & Halogen Free, RoHS/WEEE compliant

TYPICAL APPLICATIONS

- Cell phones, PDAs, MP3 players
- Digital camera, Notebooks, GPS Units
- Any battery powered equipment

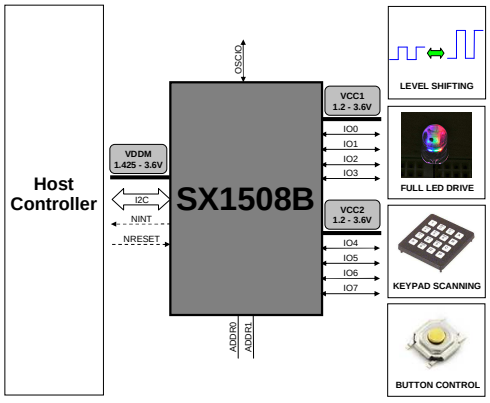


Table of Contents

GENERAL DESCRIPTION 1
ORDERING INFORMATION 1
KEY PRODUCT FEATURES 1
TYPICAL APPLICATIONS 1
1 PIN DESCRIPTION 4
1.1 SX1508B 8-channel I²C GPIO with LED Driver and Keypad Engine 4
1.2 SX1509B 16-channel I²C GPIO with LED Driver and Keypad Engine 5
1.3 I/Os Feature Summary 6
2 ELECTRICAL CHARACTERISTICS 7
2.1 Absolute Maximum Ratings 7
2.2 Electrical Specifications 7
3 TYPICAL OPERATING CHARACTERISTICS 10
4 BLOCK DETAILED DESCRIPTION 11
4.1 SX1508B 8-channel I²C GPIO with LED Driver and Keypad Engine 11
4.2 SX1509B 16-channel I²C GPIO with LED Driver and Keypad Engine 11
4.3 Reset 12
4.3.1 Hardware (NRESET) 12
4.3.2 Software (RegReset) 12
4.4 2-Wire Interface (I²C) 12
4.4.1 WRITE 13
4.4.2 READ 13
4.5 I/O Banks 14
4.5.1 Input Debouncer 14
4.5.2 Keypad Scanning Engine 14
4.5.3 Level Shifter 15
4.5.4 Polarity Inverter 16
4.6 Interrupt (NINT) 16
4.7 Clock Management 17
4.8 LED Driver 17
4.8.1 Overview 17
4.8.2 Static Mode 18
4.8.3 Single Shot Mode 18
4.8.4 Blink Mode 19
4.8.5 LED Driver Modes 19
4.8.6 Synchronization of LED Drivers across several ICs 20
4.8.7 Tutorial 20
5 CONFIGURATION REGISTERS 22
5.1 SX1508B 8-channel GPIO with LED Driver and Keypad Engine 22
5.2 SX1509B 16-channel GPIO with LED Driver and Keypad Engine 26
6 APPLICATION INFORMATION 32
6.1 Typical Application Circuit 32
6.2 Typical LED Connection 32
7 PACKAGING INFORMATION 33
7.1 QFN-UT 20-pin Outline Drawing 33
7.2 QFN-UT 20-pin Land Pattern 33

7.3 QFN-UT 28-pin Outline Drawing 34
7.4 QFN-UT 28-pin Land Pattern 34
8 SOLDERING PROFILE 35
9 MARKING INFORMATION 36

1 PIN DESCRIPTION

1.1 SX1508B 8-channel I²C GPIO with LED Driver and Keypad Engine

Pin	Symbol	Type	Description
1	NRESET	DI	Active low reset input
2	SDA	DIO	I²C serial data line
3	SCL	DI	I²C serial clock line
4	ADDR0	DI	Address input bit 0, connect to VDDM or GND
5	I/O[0]	DIO (*1)	I/O[0], at power-on configured as an input LED driver : Intensity control (PWM)
6	I/O[1]	DIO (*1)	I/O[1], at power-on configured as an input LED driver : Intensity control (PWM)
7	VCC1	P	Supply voltage for Bank A I/O[3-0]
8	GND	P	Ground Pin
9	I/O[2]	DIO (*1)	I/O[2], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
10	I/O[3]	DIO (*1)	I/O[3], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
11	NINT	DO	Active low interrupt output
12	ADDR1	DI	Address input bit 1, connect to VDDM or GND
13	OSCIO	DIO (*1)	Oscillator input/output, can also be used as GPO
14	VDDM	P	Main supply voltage
15	I/O[4]	DIO (*1)	I/O[4], at power-on configured as an input LED driver : Intensity control (PWM)
16	I/O[5]	DIO (*1)	I/O[5], at power-on configured as an input LED driver : Intensity control (PWM)
17	VCC2	P	Supply voltage for Bank B I/O[7-4]
18	GND	P	Ground Pin
19	I/O[6]	DIO (*1)	I/O[6], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
20	I/O[7]	DIO (*1)	I/O[7], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)

D/I/O/P: Digital/Input/Output/Power
(*1) This pin is programmable through the I²C interface

Table 1 – SX1508B Pin Description

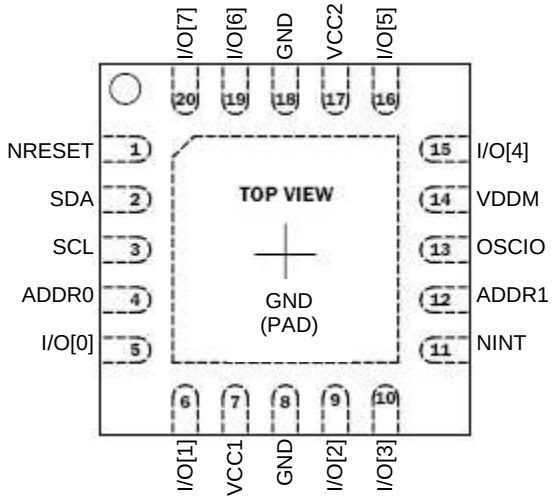


Figure 1 – SX1508B QFN-UT-20 Pinout

1.2 SX1509B 16-channel I²C GPIO with LED Driver and Keypad Engine

Pin	Symbol	Type	Description
1	I/O[2]	DIO (*1)	I/O[2], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
2	I/O[3]	DIO (*1)	I/O[3], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
3	GND	P	Ground Pin
4	VCC1	P	Supply voltage for Bank A I/O[7-0]
5	I/O[4]	DIO (*1)	I/O[4], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
6	I/O[5]	DIO (*1)	I/O[5], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
7	I/O[6]	DIO (*1)	I/O[6], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
8	I/O[7]	DIO (*1)	I/O[7], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
9	NINT	DO	Active low interrupt output
10	ADDR1	DI	Address input bit 1, connect to VDDM or GND
11	OSCIO	DIO (*1)	Oscillator input/output, can also be used as GPO
12	VDDM	P	Main supply voltage
13	I/O[8]	DIO (*1)	I/O[8], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
14	I/O[9]	DIO (*1)	I/O[9], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
15	I/O[10]	DIO (*1)	I/O[10], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
16	I/O[11]	DIO (*1)	I/O[11], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
17	GND	P	Ground Pin
18	VCC2	P	Supply voltage for Bank B I/O[15-8]
19	I/O[12]	DIO (*1)	I/O[12], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
20	I/O[13]	DIO (*1)	I/O[13], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
21	I/O[14]	DIO (*1)	I/O[14], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
22	I/O[15]	DIO (*1)	I/O[15], at power-on configured as an input LED driver : Intensity control (PWM), Blinking, Breathing (Fade In/Out)
23	NRESET	DI	Active low reset input
24	SDA	DIO	I²C serial data line
25	SCL	DI	I²C serial clock line
26	ADDR0	DI	Address input bit 0, connect to VDDM or GND
27	I/O[0]	DIO (*1)	I/O[0], at power-on configured as an input LED driver : Intensity control (PWM), Blinking
28	I/O[1]	DIO (*1)	I/O[1], at power-on configured as an input LED driver : Intensity control (PWM), Blinking

(*1) This pin is programmable through the I²C interface

Table 2 – SX1509B Pin Description

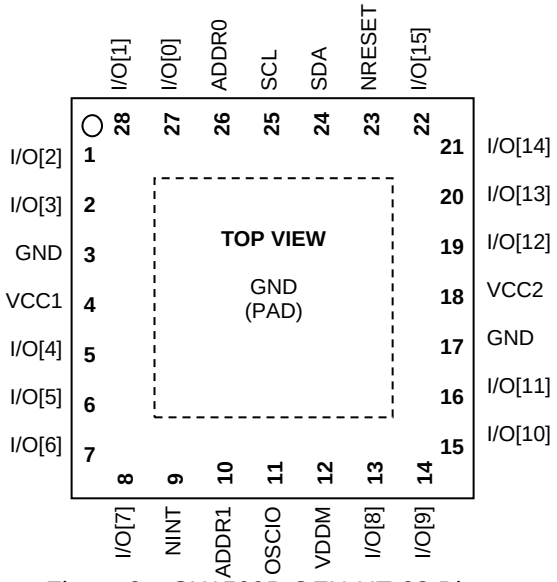


Figure 2 – SX1509B QFN-UT-28 Pinout

1.3 I/Os Feature Summary

I/O	SX1508B					SX1509B				
	LED Driver			Keypad		LED Driver			Keypad	
	PWM	Blink	Breathe	Row	Col.	PWM	Blink	Breathe	Row	Col.
0	√			√		√	√		√	
1	√			√		√	√		√	
2	√	√		√		√	√		√	
3	√	√	√	√		√	√		√	
4	√				√	√	√	√	√	
5	√				√	√	√	√	√	
6	√	√			√	√	√	√	√	
7	√	√	√		√	√	√	√	√	
8						√	√			√
9						√	√			√
10						√	√			√
11						√	√			√
12						√	√	√		√
13						√	√	√		√
14						√	√	√		√
15						√	√	√		√

Table 3 – I/Os Feature Summary

Please note that in addition to table above, all I/Os feature bank-to-bank and bank-to-host level shifting.

2 ELECTRICAL CHARACTERISTICS

2.1 Absolute Maximum Ratings

Stress above the limits listed in the following table may cause permanent failure. Exposure to absolute ratings for extended time periods may affect device reliability. The limiting values are in accordance with the Absolute Maximum Rating System (IEC 134). All voltages are referenced to ground (GND).

Symbol	Description	Min	Max	Unit
V_{max_VDDM}	Main supply voltage	- 0.4	3.7	V
V_{max_VCC1-2}	Digital I/O pin supply voltage	- 0.4	3.7	V
V_{ES_HBM}	Electrostatic handling HBM model(1) (SX1508B)	-	2000	V
	Electrostatic handling HBM model(1) (SX1509B)	-	1500	
V_{ES_CDM}	Electrostatic handling CDM model	-	1000	V
V_{ES_MM}	Electrostatic handling MM model (SX1508B)	-	200	V
	Electrostatic handling MM model (SX1509B)	-	150	
T_A	Operating ambient temperature range	-40	+85	℃
T_C	Junction temperature range	-40	+125	℃
T_{STG}	Storage temperature range	-55	+150	℃
I_{lat}	Latchup-free input pin current(2)	+/-100	-	mA

(1) Tested according to JESD22-A114A
(2) Static latch-up values are valid at maximum temperature according to JEDEC 78 specification

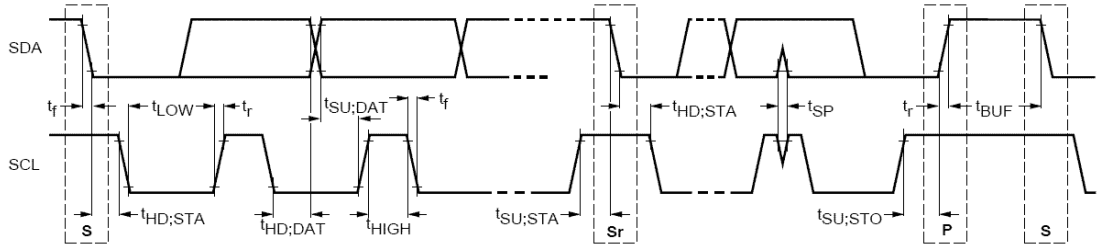
Table 4 - Absolute Maximum Ratings

2.2 Electrical Specifications

Table below assumes default registers values, unless otherwise specified. Typical values are given for T_A = +25℃, VDDM=VCC1=VCC2=3.3V.

Symbol	Description	Conditions	Min	Typ	Max	Unit
Supply						
VDDM	Main supply voltage		1.425	-	3.6	V
VCC1,2	I/O banks supply voltage		1.2	-	3.6	V
IDDM	Main supply current (SX1508B, I²C inactive)	Oscillator OFF	-	1	5	µA
		Internal osc. (2MHz)	-	175	235	
		External osc. (32kHz)	-	10	-	
	Main supply current (SX1509B, I²C inactive)	Oscillator OFF	-	1	5	µA
		Internal osc. (2MHz)	-	365	460	
		External osc. (32kHz)	-	10	-	
ICC1,2	I/O banks supply current(1)		-	1	2	µA
I/Os set as Input						
VIH	High level input voltage	VCC1,2 >= 2V	0.7* VCC1,2	-	5.5(8)	V
		VCC1,2 < 2V	0.8* VCC1,2	-	5.5(8)	
VIL	Low level input voltage	VCC1,2 >= 2V	-0.4	-	0.3* VCC1,2	V
		VCC1,2 < 2V	-0.4	-	0.2* VCC1,2	
ILEAK	Input leakage current	Assuming no active pull-up/down	-1	-	1	µA
CI	Input capacitance	-	-	-	10	pF
I/Os set as Output						
VOH	High level output voltage	-	VCC1,2 – 0.3	-	VCC1,2	V
VOL	Low level output voltage	-	-0.4	-	0.3	V
IOH	High level output source current	VCC1,2 >= 2V	-	-	8(2)	mA
		VCC1,2 < 2V	-	-	2(2)	
IOL	Low level output sink current	VCC1,2 >= 2V	-	-	15(2)	mA
		VCC1,2 < 2V	-	-	8(2)	

Symbol	Description	Conditions	Min	Typ	Max	Unit
t_{PV}	Output data valid timing	Cf. Figure 8	-	-	425	ns
NINT (Output)						
VOL	Low level output voltage	-	-0.4	-	0.3	V
IOL_M	Low level output sink current	VDDM >= 2V	-	-	8	mA
		VDDM < 2V	-	-	4	
t_{IV}	Interrupt valid timing	From input data change	-	-	4	µs
t_{IR}	Interrupt reset timing	From RegInterruptSource clearing	-	-	4	µs
NRESET (Input)						
VIH_{MR}	High level input voltage	VDDM >= 2V	0.7*VDDM	-	$VDDM_{max}$	V
		VDDM < 2V	0.8*VDDM	-	$VDDM_{max}$	
VIL_M	Low level input voltage	VDDM >= 2V	-0.4	-	0.3*VDDM	V
		VDDM < 2V	-0.4	-	0.2*VDDM	
ILEAK	Input leakage current	-	-1	-	1	µA
CI	Input capacitance	-	-	-	10	pF
VPOR	Power-On-Reset voltage	Cf. Figure 6	-	0.8	-	V
VDROPH	High brown-out voltage	Cf. Figure 6	-	VDDM-1	-	V
VDROPL	Low brown-out voltage	Cf. Figure 6	-	0.2	-	V
t_{RESET}	Reset time	Cf. Figure 6	-	-	2.5	ms
t_{PULSE}	Reset pulse from host uC	Cf. Figure 6	200	-	-	ns
ADDR0, ADDR1 (Inputs)						
VIH_{MA}	High level input voltage	VDDM >= 2V	0.7*VDDM	-	VDDM+0.3	V
		VDDM < 2V	0.8*VDDM	-	VDDM+0.3	
VIL_M	Low level input voltage	VDDM >= 2V	-0.4	-	0.3*VDDM	V
		VDDM < 2V	-0.4	-	0.2*VDDM	
ILEAK	Input leakage current	-	-1	-	1	µA
CI	Input capacitance	-	-	-	10	pF
OSCIO (Input/Output)						
VIH_{MO}	High level input voltage	VDDM >= 2V	0.7*VDDM	-	VDDM+0.3	V
		1.425V =< VDDM < 2V	0.8*VDDM	-	VDDM+0.3	
		VDDM < 1.425V	0.9*VDDM	-	VDDM+0.3	
VIL_{MO}	Low level input voltage	VDDM >= 2V	-0.4	-	0.3*VDDM	V
		1.425V =< VDDM < 2V	-0.4	-	0.2*VDDM	
		VDDM < 1.425V	-0.4	-	0.1*VDDM	
ILEAK	Input leakage current	-	-1	-	1	µA
CI	Input capacitance	-	-	-	10	pF
VOH_M	High level output voltage	-	VDDM-0.3	-	VDDM	V
VOL	Low level output voltage	-	-0.4	-	0.3	V
IOH_M	High level output source current	VDDM >= 2V	-	-	8	mA
		VDDM < 2V	-	-	2	
IOL_M	Low level output sink current	VDDM >= 2V	-	-	8	mA
		VDDM < 2V	-	-	4	
SCL (Input) and SDA (Input/Output) (3)						
Interface complies with slave F/S mode I^2C interface as described by Philips I^2C specification version 2.1 dated January, 2000. Please refer to that document for more detailed I^2C specifications.						
VOL	Low level output voltage	-	-0.4	-	0.3	V
IOL_M	Low level output sink current	VDDM >= 2V	-	-	8	mA
		VDDM < 2V	-	-	4	
VIH_{MR}	High level input voltage	VDDM >= 2V	0.7*VDDM	-	$VDDM_{max}$	V
		VDDM < 2V	0.8*VDDM	-	$VDDM_{max}$	
VIL_M	Low level input voltage	VDDM >= 2V	-0.4	-	0.3*VDDM	V
		VDDM < 2V	-0.4	-	0.2*VDDM	



Symbol	Description	Conditions	Min	Typ	Max	Unit
f_{SCL}	SCL clock frequency	-	-	-	400	kHz
$t_{HD;STA}$	Hold time (repeated) START condition	-	0.6	-	-	µs
t_{LOW}	LOW period of the SCL clock	-	1.3	-	-	µs
t_{HIGH}	HIGH period of the SCL clock	-	0.6	-	-	µs
$t_{SU;STA}$	Set-up time for a repeated START condition	-	0.6	-	-	µs
$t_{HD;DAT}$	Data hold time	-	0(4)	-	0.9(5)	µs
$t_{SU;DAT}$	Data set-up time	-	100(6)	-	-	ns
t_r	Rise time of both SDA and SCL	-	20+0.1C_b(7)	-	300	ns
t_f	Fall time of both SDA and SCL	-	20+0.1C_b(7)	-	300	ns
$t_{SU;STO}$	Set-up time for STOP condition	-	0.6	-	-	µs
t_{BUF}	Bus free time between a STOP and START condition	-	1.3	-	-	µs
C_b	Capacitive load for each bus line	-	-	-	400	pF
V_{nL}	Noise margin at the LOW level for each connected device (including hysteresis)	-	-	0.1* VDDM	-	V
V_{nH}	Noise margin at the HIGH level for each connected device (including hysteresis)	-	-	0.2* VDDM	-	V
t_{SP}	Pulse width of spikes suppressed by the input filter	-	-	-	50	ns
Miscellaneous						
RPULL	Programmable pull-up/down resistors for IO[0-7]	-	-	42	-	kΩ
f_{OSC}	Oscillator frequency	Internal	1.3	2	2.6	MHz
		External from OSCIN (40-60% duty cycle)	-	-	2.6	

(1) Assuming no load connected to outputs and inputs fixed to VCC1,2 or GND.
(2) Can be increased by tying together and driving simultaneously several I/Os.
(3) All values referred to $VIH_{MR\ min}$ and $VIL_{M\ max}$ levels.
(4) A device must internally provide a hold time of at least 300ns for the SDA signal (referred to $VIH_{MR\ min}$) to bridge the undefined region of the falling edge of SCL.
(5) The maximum $t_{HD;DAT}$ has only to be met if the device does not stretch the LOW period (t_{LOW}) of the SCL signal.
(6) A Fast-mode I^2C-bus device can be used in a Standard-mode I^2C-bus system, but the requirement $t_{SU;DAT} \geq 250$ ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the SCL signal.
If such a device does stretch the LOW period of the SCL signal, it must output the next data bit to the SDA line $t_{r\ max} + t_{SU;DAT} = 1000 + 250 = 1250$ ns (according to the Standard-mode I2C-bus specification) before the SCL line is released.
(7) C_b = total capacitance of one bus line in pF. If mixed with Hs-mode devices, faster fall-times are allowed.
(8) With RegHighInput bit enabled (VCCx min =1.65V), else 3.6V (VCCx min = 1.2V)

Table 5 – Electrical Specifications

3 TYPICAL OPERATING CHARACTERISTICS

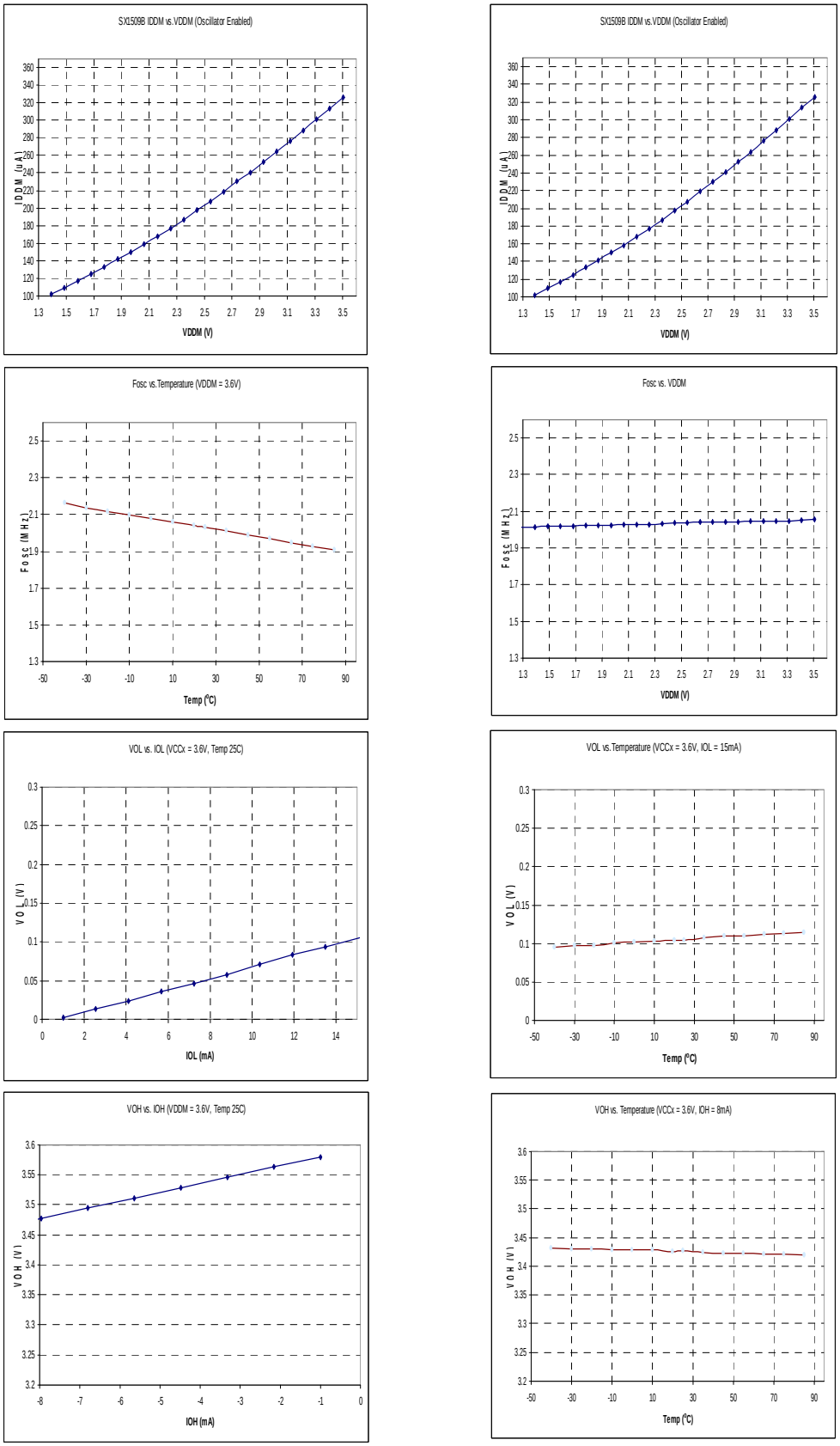


Figure 3 – Typical Operating Characteristics

4 BLOCK DETAILED DESCRIPTION

4.1 SX1508B 8-channel I²C GPIO with LED Driver and Keypad Engine

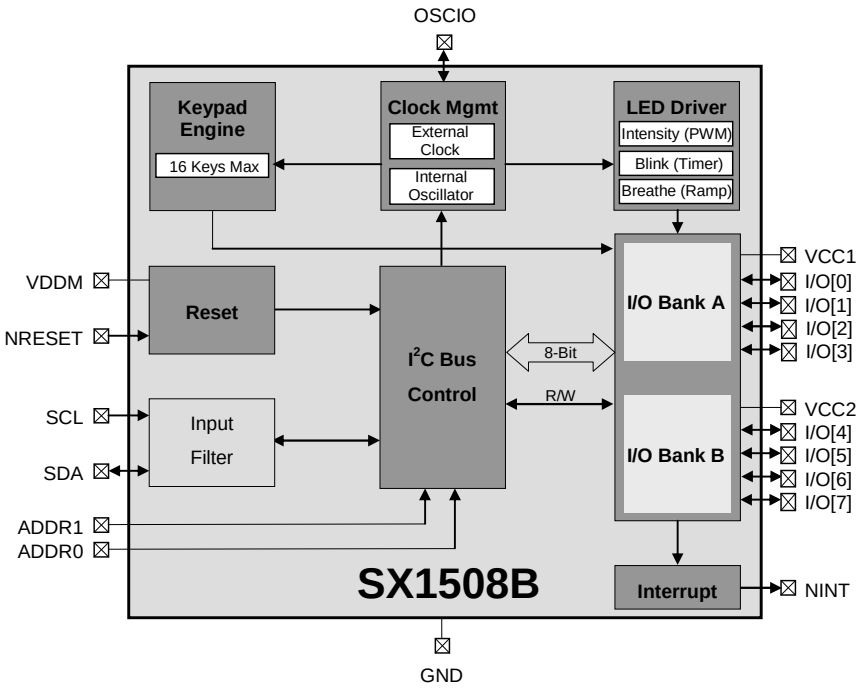


Figure 4 – 8-channel Low Voltage GPIO with LED Driver and Keypad Engine

4.2 SX1509B 16-channel I²C GPIO with LED Driver and Keypad Engine

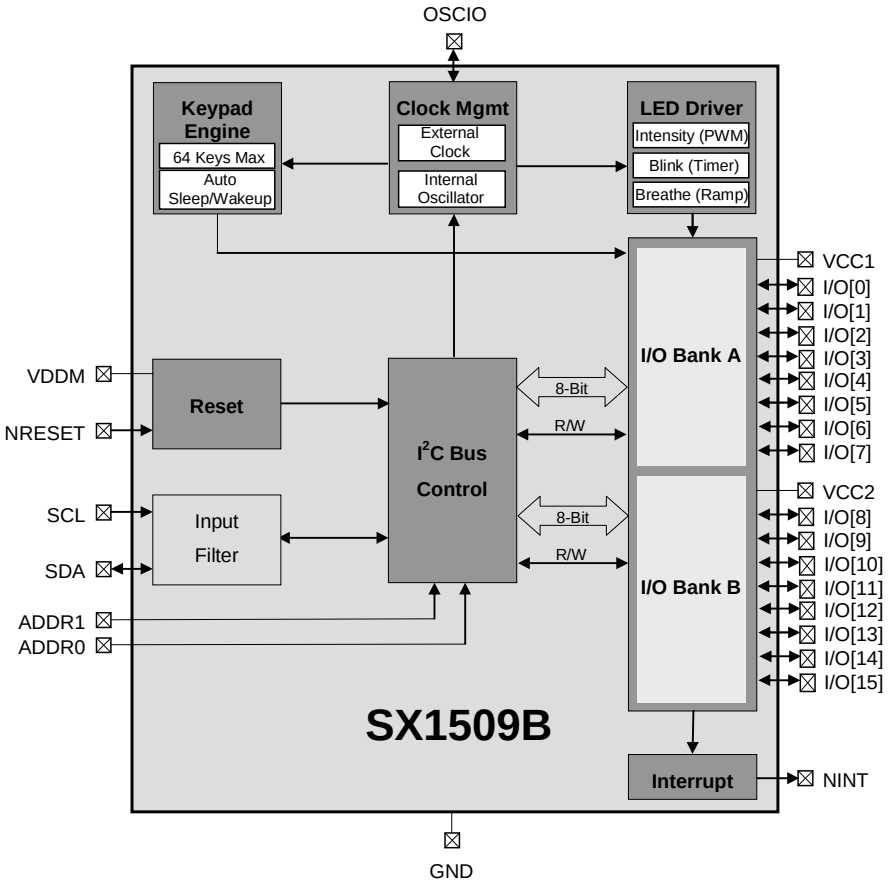


Figure 5 – 16-channel Low Voltage GPIO with LED Driver and Keypad Engine

4.3 Reset

4.3.1 Hardware (NRESET)

The SX1508B and SX1509B generate their own power on reset signal after a power supply is connected to the VDDM pin. NRESET input pin can be used to reset the chip anytime, it must be connected to VDDM (or greater) either directly (if not used), or via a resistor.

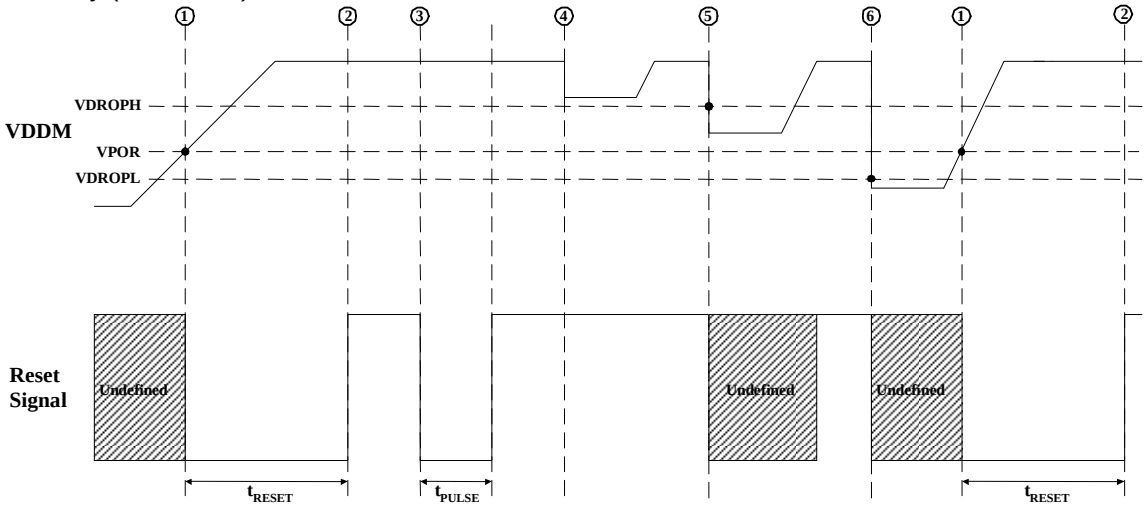


Figure 6 – Power-On / Brown-out Reset Conditions

1. Device behavior is undefined until VDDM rises above VPOR, at which point internal reset procedure is started.
2. After t_{RESET}, the reset procedure is completed.
3. In operation, the SX1508B and SX1509B may be reset (POR like or LED driver counters only depending on RegMisc setting) at anytime by an external device driving NRESET low for t_{PULSE} or longer. Chip can be accessed normally again after NRESET rising edge.
4. During a brown-out event, if VDDM drops above VDROPH a reset will not occur.
5. During a brown-out event, if VDDM drops between VDROPH and VDROPL a reset may occur.
6. During a brown-out event, if VDDM drops below VDROPL a reset will occur next time VPOR is crossed.

Please note that a brown-out event is defined as a transient event on VDDM. If VDDM is attached to a battery, then the gradual decay of the battery voltage will not be interpreted as a brown-out event.
Please also note that a sharp rise in VDDM (> 1V/us) may induce a circuit reset.

4.3.2 Software (RegReset)

Writing consecutively 0x12 and 0x34 to RegReset register will reset all registers to their default values.

4.4 2-Wire Interface (I^2C)

The SX1508B and SX1509B 2-wire interface operates only in slave mode. In this configuration, the device has one or 4 possible devices addresses defined by ADDR[1:0] pins:

Device	ADDR[1:0]	Address	Description
SX1508B	**00**	0x2**0** (0100**00**)	First address of the 2-wire interface
	01	0x2**1** (0100**01**)	Second address of the 2-wire interface
	10	0x2**2** (0100**10**)	Third address of the 2-wire interface
	11	0x2**3** (0100**11**)	Fourth address of the 2-wire interface
SX1509B	**00**	0x3E (**0**111110)	First address of the 2-wire interface
	01	0x3F (**0**111111)	Second address of the 2-wire interface
	10	0x70 (**1**110000)	Third address of the 2-wire interface
	11	0x71 (**1**110001)	Fourth address of the 2-wire interface

Table 6 - 2-Wire Interface Address

2 lines are used to exchange data between an external master host and the slave device:

- **SCL** : **S**erial **CL**ock
- **SDA** : **S**erial **DA**ta

The SX1508B and SX1509B are read-write slave-mode I^2C devices and comply with the Philips I^2C standard Version 2.1 dated January, 2000. The SX1508B and SX1509B have a few user-accessible internal 8-bits registers to set the various parameters of operation (Cf. §5 for detailed configuration registers description). The I^2C interface has been designed for program flexibility, in that once the slave address has been sent to the SX1508B or SX1509B enabling it to be a slave transmitter/receiver, any register can be written or read independently of each other. The start and stop commands frame the data-packet and the repeat start condition is allowed if necessary.

Seven bit addressing is used and ten bit addressing is not allowed. Any general call address will be ignored by the SX1508B and SX1509B. The SX1508B and SX1509B are not CBUS compatible and can operate in standard mode (100kbit/s) or fast mode (400kbit/s).

4.4.1 WRITE

After the start condition [S], the slave address (SA) is sent, followed by an eighth bit ('0') indicating a Write. The slave then Acknowledges [A] that it is being addressed, and the Master sends an 8 bit Data Byte consisting of the slave Register Address (RA). The Slave Acknowledges [A] and the master sends the appropriate 8 bit Data Byte (WD0). Again the slave Acknowledges [A]. In case the master needs to write more data, a succeeding 8 bit Data Byte will follow (WD1), acknowledged by the slave [A]. This sequence will be repeated until the master terminates the transfer with the Stop condition [P].

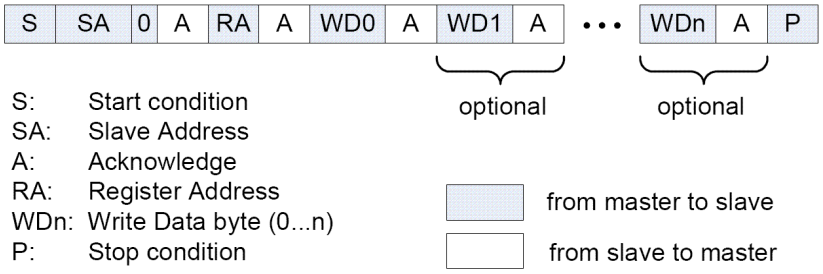


Figure 7 - 2-Wire Serial Interface, Write Operation

When successive register data (WD1...WDn) is supplied by the master, the register address can be automatically incremented or kept fixed depending on the setting programmed in RegMisc.

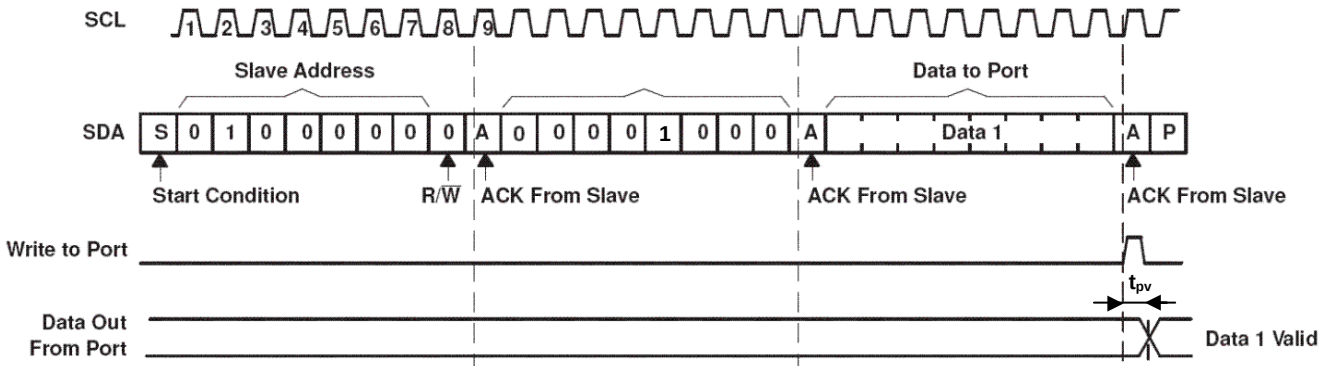


Figure 8 – Example: Write RegData Register

4.4.2 READ

After the start condition [S], the slave address (SA) is sent, followed by an eighth bit ('0') indicating a Write. The slave then Acknowledges [A] that it is being addressed, and the Master responds with an 8 bit Data consisting of the Register Address (RA). The slave Acknowledges [A] and the master sends the Repeated Start Condition [Sr]. Once again, the slave address (SA) is sent, followed by an eighth bit ('1') indicating a Read.

The slave responds with an Acknowledge [A] and the read Data byte (RD0). If the master needs to read more data it will acknowledge [A] and the slave will send the next read byte (RD1). This sequence can be repeated until the master terminates with a NACK [N] followed by a stop [P].

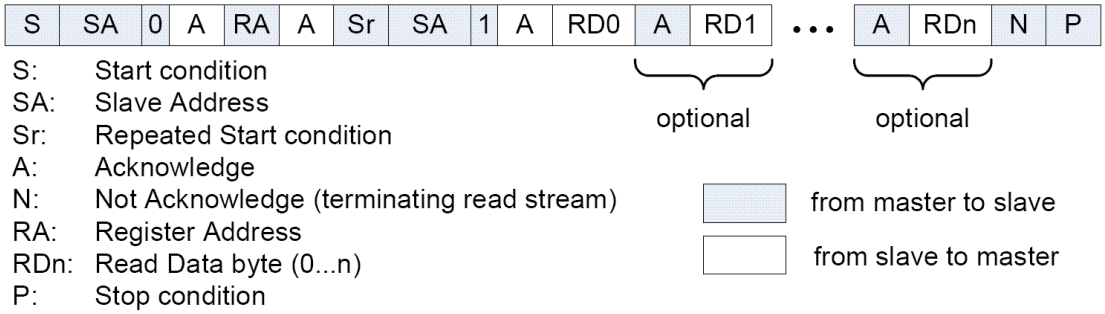


Figure 9 - 2-Wire Serial Interface, Read Operation

When successive register data (RD1...RDn) is read by the master, the register address will be automatically incremented or kept fixed depending on the setting programmed in RegMisc.

4.5 I/O Banks

4.5.1 Input Debouncer

Each input can be individually debounced by setting corresponding bits in RegDebounce register. At power up the debounce function is disabled. After enabling the debouncer, the change of the input value is accepted only if the input value is identical at two consecutive sampling times.
The debounce time common to all IOs can be set in RegDebounceConfig register from 0.5 to 64ms (fOSC = 2MHz).

4.5.2 Keypad Scanning Engine

SX1508B, and SX1509B integrate a fully programmable keypad scanning engine to implement keypad applications up to 8x8 matrix (i.e. 64 keys).
Please note that SX1509B also implements an Auto Sleep/Wakeup feature to save power consumption when no key has been pressed for a programmed time.

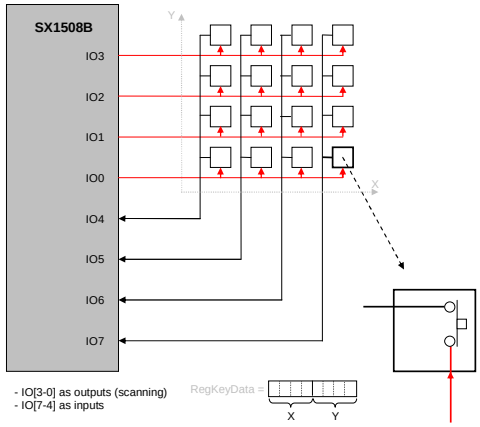


Figure 10 – 4x4 Keypad Connection to SX1508B

Following procedure should be implemented on the host controller for a 4x4 keypad:

1. Set RegDir to 0xF0 (IO[3-0] as outputs, IO[7-4] as inputs) , set RegOpenDrain to 0x0F (IO[3-0] as open-drain outputs), set RegPullup to 0xF0 (pull-ups enabled on inputs IO[7-4]).

2. Enable and configure debouncing on IO[7-4] (RegDebounceEnable = 0xF0, Ex : RegDebounceConfig = 0x05)

3. Enable and configure keypad scanning engine (Ex : RegKeyConfig = 0x7D) This will start an infinite loop with the following sequence to IO[3:0]: ZZZ0, ZZ0Z, Z0ZZ, 0ZZZ. Make sure that scan interval is set to higher value than the debounce time.

4. When a key is pressed, NINT goes low, key scan is halted and the key coordinates are stored in RegKeyData:

- *The column data will be stored in RegKeyData[7:4] (Note: column indication is active low)*

- *The row data will be stored in RegKeyData[3:0] (Note: row indication is active low)*
- *When RegKeyData is read, this data along with the interrupt is automatically cleared (same behavior as reading RegData) and the key scan continues to the next row.*

5. Restart from point 4.

This implementation allows the host to handle both single and multi-touches easily (fast AAAAAA sequence is a long press of key A, fast ABABABAB sequence is key A and key B pressed together, etc)

4.5.3 Level Shifter

Because of their 5.5V tolerant I/O banks with independent supply voltages between 1.2V and 3.6V, the SX1508B and SX1509B can perform level shifting of signals from one I/O bank to another **without uC activity** by programming the corresponding configuration register bits accordingly in RegLevelShifter (and RegDir).
This can save significant BOM cost in a final application where only a few signals need to be level-shifted (no need for an additional external level shifter IC).

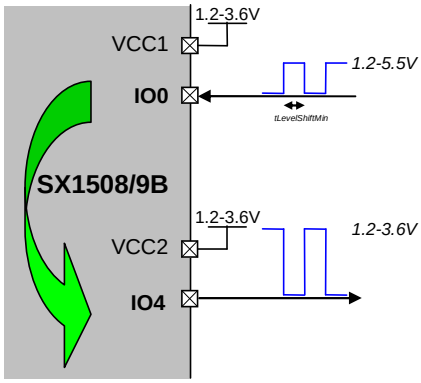


Figure 11 – Level Shifting Example

The minimum pulse width tLevelShiftMin which can be level shifted properly depends on VCCx and VDDM:

tLevelShiftMin = Input Delay + Core Delay + Output Delay

Input/Core/Output delays vs VCCx/VDDM are given in figures below.

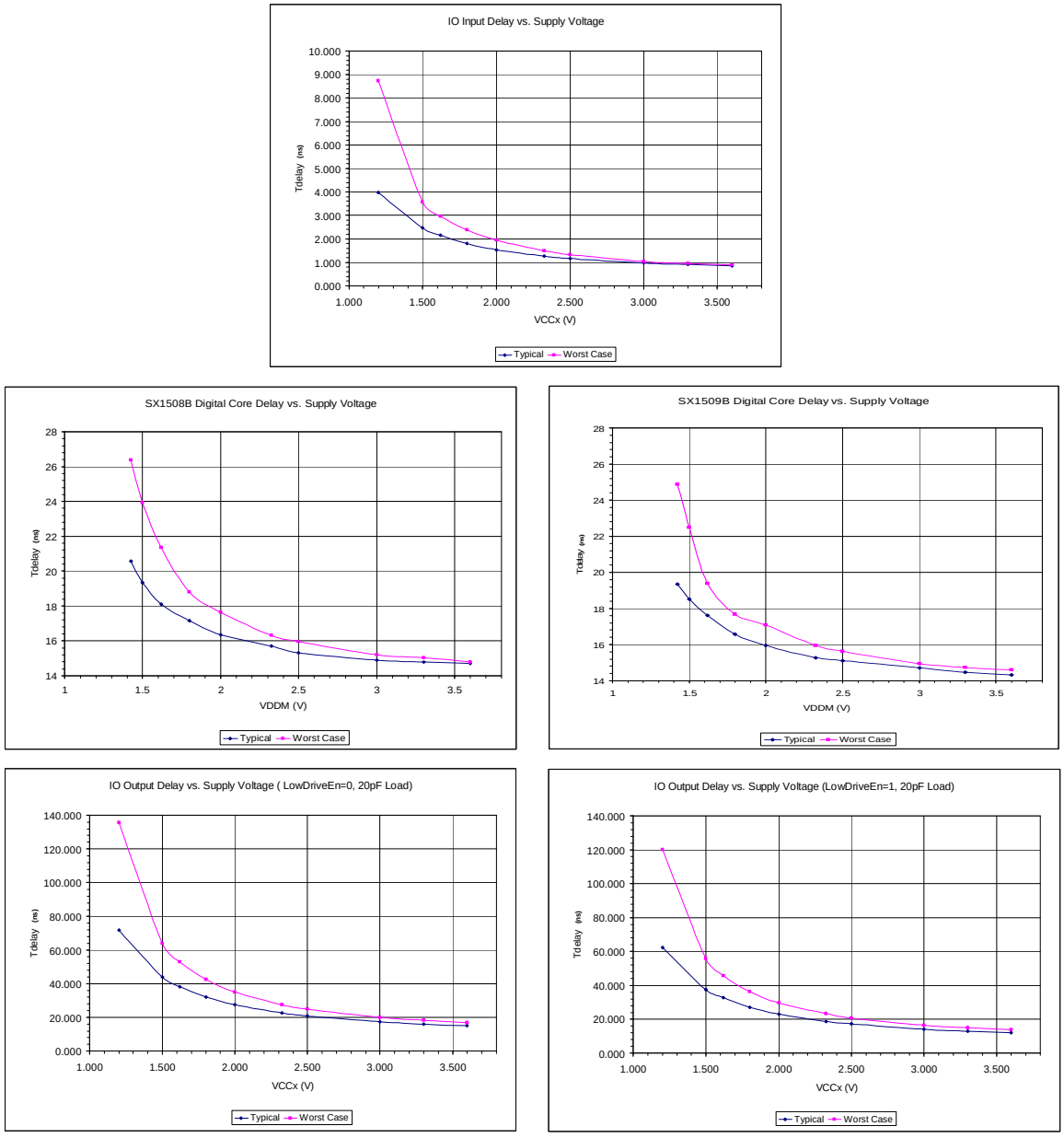


Figure 12 – Level Shifter Max Frequency Calculation Data

4.5.4 Polarity Inverter

Each IO's polarity can be individually inverted by setting corresponding bit in RegPolarity register. Please note that polarity inversion can also be combined with level shifting feature.

4.6 Interrupt (NINT)

At start-up, the transition detection logic is reset, and NINT is released to a high-impedance state. The interrupt mask register is set to 0xFF, disabling the interrupt output for transitions on all I/O ports. The transition flags are cleared to indicate no data changes.

An interrupt NINT can be generated on any programmed combination of I/Os rising and/or falling edges through the RegInterruptMask and RegSense registers.
If needed, the I/Os which triggered the interrupt can then be identified by reading RegInterruptSource register.

When NINT is low (i.e. interrupt occurred), it can be reset back high (i.e. cleared) by writing 0xFF in RegInterruptSource (this will also clear corresponding bits in RegEventStatus register).
The interrupt can also be cleared automatically when reading RegData register (Cf. RegMisc)

Example: We want to detect rising edge of I/O[1] on SX1508B (NINT will go low).

1. *We enable interrupt on I/O[1] in RegInterruptMask*

⇨ *RegInterruptMask ="XXXXXX**0**X"*

2. We set edge sense for I/O[1] in RegSense
⇨ *RegSenseLow ="XXXX**01**XX"*

Please note that independently from the "user defined" process described above the keypad engine, when enabled, also uses NINT to indicate a key press.
Hence we have NINT = "user defined condition occurred" OR "keypad engine condition occurred".

4.7 Clock Management

A main oscillator clock fOSC is needed by the LED driver, keypad engine and debounce features.

Clock management block is illustrated in figure below.

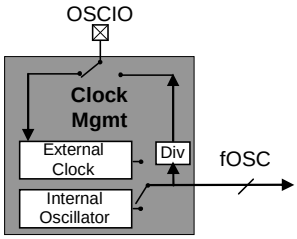


Figure 13 – Clock Management Overview

The block is configured in register RegClock (Cf §5 for more detailed information):
- Selection of internal clock source: none (OFF) or internal oscillator or external clock input from OSCIN.
- Definition of OSCIO pin function (OSCIN or OSCOUT)
- OSCOUT frequency setting (sub-multiple of fOSC)

Please note that if needed the OSCOUT feature can be used as an additional GPO (Cf. RegClock)

4.8 LED Driver

4.8.1 Overview

Every IO has its own independent LED driver (Cf §6.2 for typical LED connection) , all IOs can perform intensity control (PWM) while some of them additionally include blinking and breathing features (Cf pin description §1)

The LED drivers of all I/Os share the same clock ClkX configurable in RegMisc[6:4]. Please note that for power consumption reasons ClkX is OFF by default.

Assuming ClkX is not OFF, LED driver for IO[X] is enabled when RegLEDDriverEnable[X] = 1 in which case it can operate in one of the three modes below:

- Static mode (all I/Os, with or without fade in/out)
- Single shot mode (blinking capable I/Os only, with or without fade in/out)
- Blink mode (blinking capable I/Os only, with or without fade in/out)

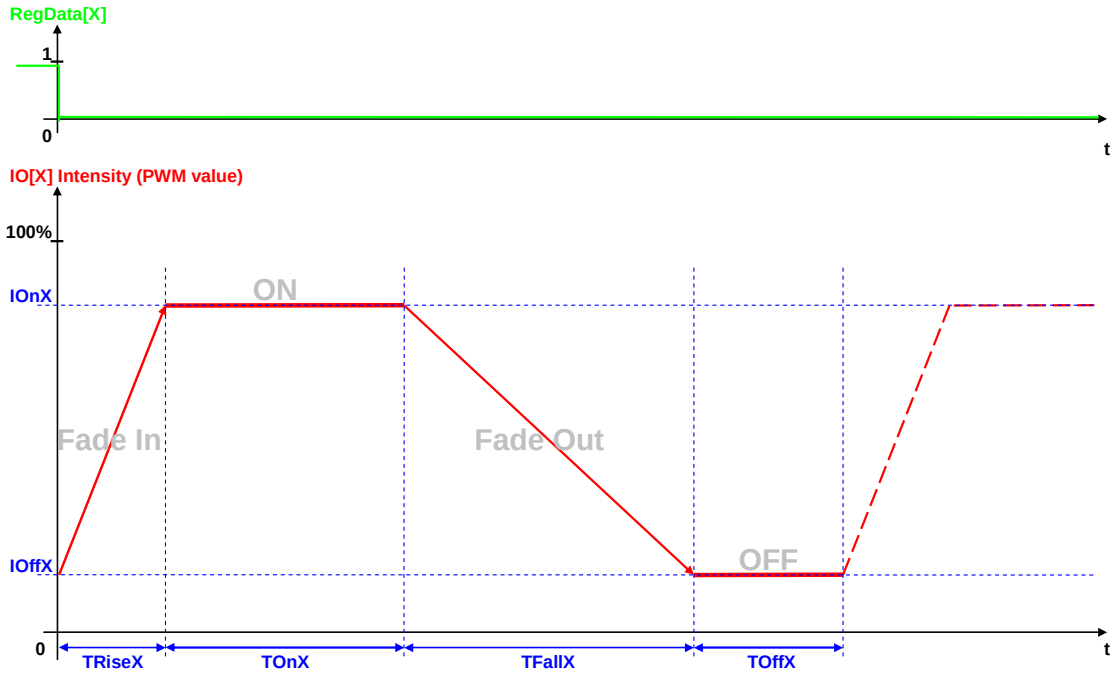


Figure 14 – LED Driver Overview

Each IO[X] has its own set of programmable registers (Cf §5 for more detailed information):

- **RegTOnX** (blinking capable I/Os only): TOnX, ON time of IO[X]
- **RegIOnX** (all I/Os): IOnX, ON intensity of IO[X]
- **RegOffX** (blinking capable I/Os only): TOffX and IOffX, OFF time and intensity of IO[X]
- **RegTRiseX**(breathing capable I/Os only): TRiseX, fade in time of IO[X]
- **RegTFallX**(breathing capable I/Os only): TFallX, fade out time of IO[X]

Please note that the LED driver mode is selectable for each IO bank between linear and logarithmic. (Cf §4.8.5)

All the figures assume normal IO polarity, for inverse polarity RegData control must be inverted (does not invert the polarity of the IO signal itself).

4.8.2 Static Mode

Only mode available for non blinking capable IOs (with Off intensity = 0), else invoked when TOnX = 0.
If the I/O doesn't support fading the LED intensity will step directly to the IOnX/IOffX value.



Figure 15 – LED Driver Static Mode

4.8.3 Single Shot Mode

Invoked when TOnX != 0 and TOffX = 0.
If the I/O doesn't support fading the LED intensity will step directly to the IOnX/IOffX value.

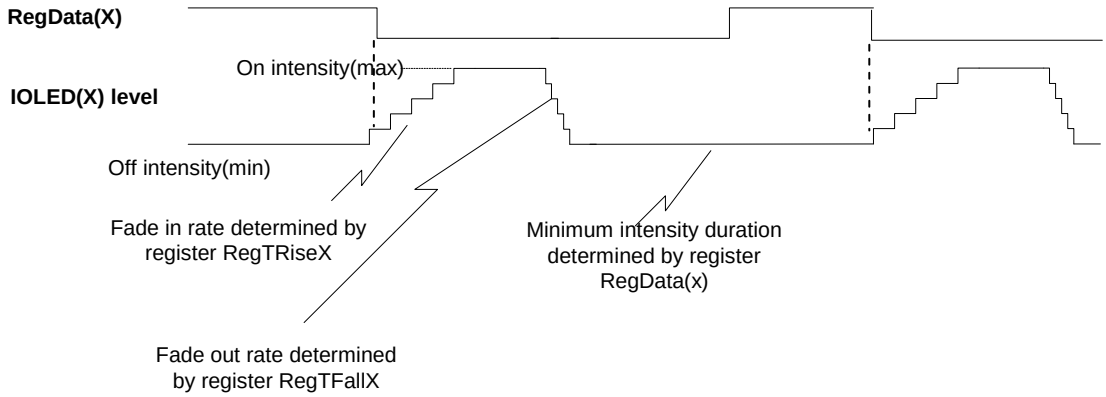


Figure 16 – LED Driver Single Shot Mode

4.8.4 Blink Mode

Invoked when TOnX != 0 and TOffX != 0.
If the I/O doesn't support fading the LED intensity will step directly to the IOnX/IOffX value.

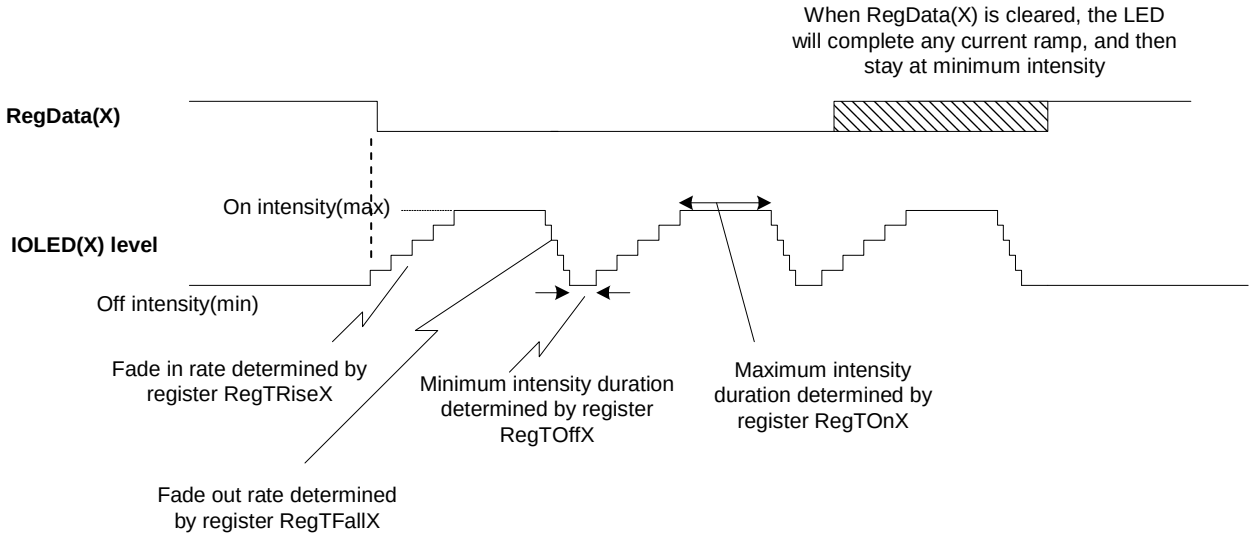


Figure 17 – LED Driver Blink Mode

4.8.5 LED Driver Modes

For each IO bank, the LED driver mode of fading capable IOs can be selected between linear or logarithmic in RegMisc.

Lin.	Log.	Lin.	Log.	Lin.	Log.	Lin.	Log.	Lin.	Log.	Lin.	Log.	Lin.	Log.	Lin.	Log.
0	0	32	4	64	13	96	28	128	53	160	88	192	135	224	198
1	0	33	4	65	13	97	28	129	53	161	88	193	135	225	198
2	0	34	4	66	13	98	30	130	53	162	88	194	135	226	198
3	0	35	4	67	13	99	30	131	53	163	88	195	135	227	198
4	0	36	5	68	14	100	31	132	56	164	93	196	142	228	207
5	0	37	5	69	14	101	31	133	56	165	93	197	142	229	207
6	0	38	5	70	14	102	32	134	56	166	93	198	142	230	207
7	0	39	5	71	14	103	32	135	56	167	93	199	142	231	207
8	1	40	6	72	16	104	34	136	60	168	98	200	150	232	216
9	1	41	6	73	16	105	34	137	60	169	98	201	150	233	216
10	1	42	6	74	17	106	35	138	60	170	98	202	150	234	216
11	1	43	6	75	17	107	35	139	60	171	98	203	150	235	216
12	1	44	7	76	18	108	36	140	65	172	104	204	157	236	225

13	1	45	7	77	18	109	36	141	65	173	104	205	157	237	225
14	1	46	7	78	19	110	38	142	65	174	104	206	157	238	225
15	1	47	7	79	19	111	38	143	65	175	104	207	157	239	225
16	2	48	8	80	20	112	39	144	69	176	110	208	165	240	235
17	2	49	8	81	20	113	39	145	69	177	110	209	165	241	235
18	2	50	8	82	21	114	41	146	69	178	110	210	165	242	235
19	2	51	8	83	21	115	41	147	69	179	110	211	165	243	235
20	2	52	9	84	22	116	42	148	73	180	116	212	172	244	245
21	2	53	9	85	22	117	42	149	73	181	116	213	172	245	245
22	2	54	9	86	23	118	44	150	73	182	116	214	172	246	245
23	2	55	9	87	23	119	44	151	73	183	116	215	172	247	245
24	3	56	10	88	24	120	46	152	78	184	122	216	181	248	255
25	3	57	10	89	24	121	46	153	78	185	122	217	181	249	255
26	3	58	10	90	25	122	46	154	78	186	122	218	181	250	255
27	3	59	10	91	25	123	46	155	78	187	122	219	181	251	255
28	3	60	11	92	26	124	49	156	83	188	129	220	189	252	255
29	3	61	11	93	26	125	49	157	83	189	129	221	189	253	255
30	3	62	12	94	27	126	49	158	83	190	129	222	189	254	255
31	3	63	12	95	27	127	49	159	83	191	129	223	189	255	255

Table 7 – LED Driver Linear vs Logarithmic Function (I)

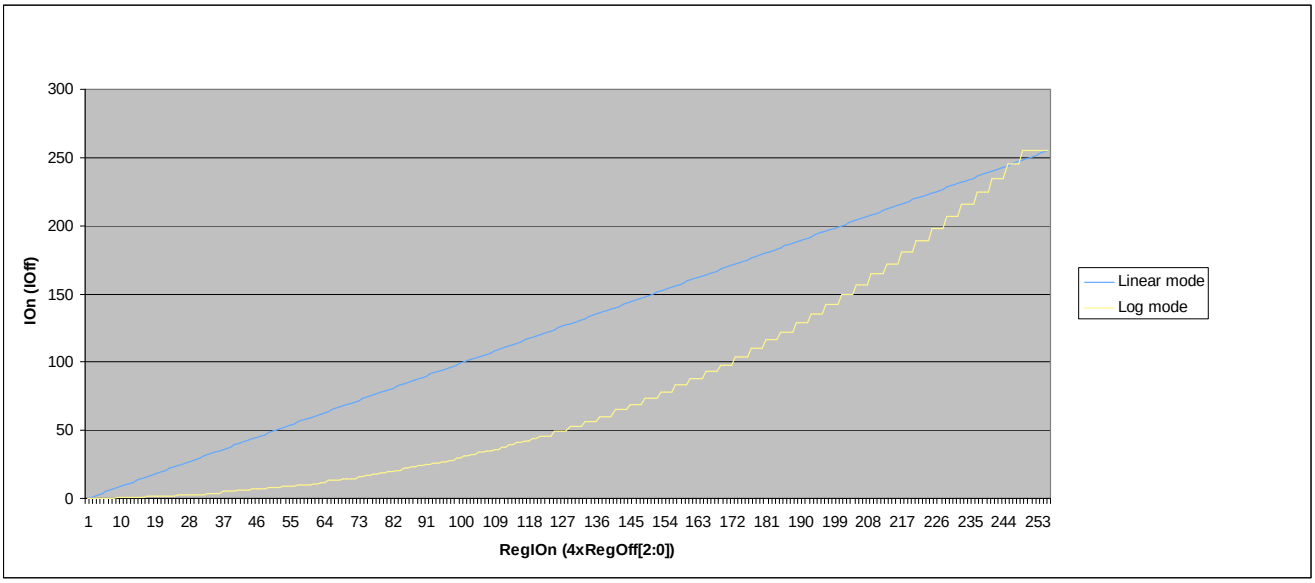


Figure 18 – LED Driver Linear vs Logarithmic Function (II)

4.8.6 Synchronization of LED Drivers across several ICs

When several GPIO expanders are used in the same application it may be useful that their LEDs drivers are synchronous for coherent global operation.

In this case all ICs should share their fOSC through their OSCIO pins and have their reset connected together.

When RegMisc of each IC is set accordingly, NRESET signal can then be used to reset all devices' internal counters (but not the register settings) and allow synchronous LED operation (blinking, fading) across multiple devices.

4.8.7 Tutorial

Below are the steps required to use the LED driver with the typical LED connection described §6.2:

- Disable input buffer (RegInputDisable)
- Disable pull-up (RegPullUp)

- Enable open drain (RegOpenDrain)
- Set direction to output (RegDir) – by default RegData is set high => LED OFF
- Enable oscillator (RegClock)
- Configure LED driver clock and mode if relevant (RegMisc)
- Enable LED driver operation (RegLEDDriverEnable)
- Configure LED driver parameters (RegTOn, RegIOn, RegOff, RegTRise, RegTFall)
- Set RegData bit low => LED driver started

5 CONFIGURATION REGISTERS

5.1 SX1508B 8-channel GPIO with LED Driver and Keypad Engine

Address	Name	Description	Default
Device and IO Banks			
0x00	**RegInputDisable**	Input buffer disable register	0000 0000
0x01	**RegLongSlew**	Output buffer long slew register	0000 0000
0x02	**RegLowDrive**	Output buffer low drive register	0000 0000
0x03	**RegPullUp**	Pull-up register	0000 0000
0x04	**RegPullDown**	Pull-down register	0000 0000
0x05	**RegOpenDrain**	Open drain register	0000 0000
0x06	**RegPolarity**	Polarity register	0000 0000
0x07	**RegDir**	Direction register	1111 1111
0x08	**RegData**	Data register	1111 1111*
0x09	**RegInterruptMask**	Interrupt mask register	1111 1111
0x0A	**RegSenseHigh**	Sense register for I/O[7:4]	0000 0000
0x0B	**RegSenseLow**	Sense register for I/O[3:0]	0000 0000
0x0C	**RegInterruptSource**	Interrupt source register	0000 0000
0x0D	**RegEventStatus**	Event status register	0000 0000
0x0E	**RegLevelShifter**	Level shifter register	0000 0000
0x0F	**RegClock**	Clock management register	0000 0000
0x10	**RegMisc**	Miscellaneous device settings register	0000 0000
0x11	**RegLEDDriverEnable**	LED driver enable register	0000 0000
Debounce and Keypad Engine			
0x12	**RegDebounceConfig**	Debounce configuration register	0000 0000
0x13	**RegDebounceEnable**	Debounce enable register	0000 0000
0x14	**RegKeyConfig**	Key scan configuration register	0000 0000
0x15	**RegKeyData**	Key value	1111 1111
LED Driver (PWM, blinking, breathing)			
0x16	**RegIOn0**	ON intensity register for I/O[0]	1111 1111
0x17	**RegIOn1**	ON intensity register for I/O[1]	1111 1111
0x18	**RegTOn2**	ON time register for I/O[2]	0000 0000
0x19	**RegIOn2**	ON intensity register for I/O[2]	1111 1111
0x1A	**RegOff2**	OFF time/intensity register for I/O[2]	0000 0000
0x1B	**RegTOn3**	ON time register for I/O[3]	0000 0000
0x1C	**RegIOn3**	ON intensity register for I/O[3]	1111 1111
0x1D	**RegOff3**	OFF time/intensity register for I/O[3]	0000 0000
0x1E	**RegTRise3**	Fade in register for I/O[3]	0000 0000
0x1F	**RegTFall3**	Fade out register for I/O[3]	0000 0000
0x20	**RegIOn4**	ON intensity register for I/O[4]	1111 1111
0x21	**RegIOn5**	ON intensity register for I/O[5]	1111 1111
0x22	**RegTOn6**	ON time register for I/O[6]	0000 0000
0x23	**RegIOn6**	ON intensity register for I/O[6]	1111 1111
0x24	**RegOff6**	OFF time/intensity register for I/O[6]	0000 0000
0x25	**RegTOn7**	ON time register for I/O[7]	0000 0000
0x26	**RegIOn7**	ON intensity register for I/O[7]	1111 1111
0x27	**RegOff7**	OFF time/intensity register for I/O[7]	0000 0000
0x28	**RegTRise7**	Fade in register for I/O[7]	0000 0000
0x29	**RegTFall7**	Fade out register for I/O[7]	0000 0000
Miscellaneous			
0x2A	**RegHighInput**	High input enable register	0000 0000
Software Reset			
0x7D	**RegReset**	Software reset register	0000 0000
Test (not to be written)			
0x7E	**RegTest1**	Test register	0000 0000
0x7F	**RegTest2**	Test register	0000 0000

*Bits set as output take "1" as default value.

Table 8 – SX1508B Configuration Registers Overview

Addr	Name	Default	Bits	Description	
0x00	**RegInputDisable**	0x00	7:0	Disables the input buffer of each IO 0 : Input buffer is enabled (input actually being used) 1 : Input buffer is disabled (input actually not being used or LED connection)	
0x01	**RegLongSlew**	0x00	7:0	Enables increased slew rate of the output buffer of each [output-configured] IO 0 : Increased slew rate is disabled 1 : Increased slew rate is enabled	
0x02	**RegLowDrive**	0x00	7:0	Enables reduced drive of the output buffer of each [output-configured] IO 0 : Reduced drive is disabled 1 : Reduced drive is enabled. IOL specifications are divided by 2.	
0x03	**RegPullUp**	0x00	7:0	Enables the pull-up for each IO 0 : Pull-up is disabled 1 : Pull-up is enabled	
0x04	**RegPullDown**	0x00	7:0	Enables the pull-down for each IO 0 : Pull-down is disabled 1 : Pull-down is enabled	
0x05	**RegOpenDrain**	0x00	7:0	Enables open drain operation for each [output-configured] IO 0 : Regular push-pull operation 1 : Open drain operation	
0x06	**RegPolarity**	0x00	7:0	Enables polarity inversion for each IO 0 : Normal polarity : RegData[x] = IO[x] 1 : Inverted polarity : RegData[x] = !IO[x] (for both input and output configured IOs)	
0x07	**RegDir**	0xFF	7:0	Configures direction for each IO. 0 : IO is configured as an output 1 : IO is configured as an input	
0x08	**RegData**	0xFF	7:0	Write: Data to be output to the output-configured IOs Read: Data seen at the IOs, independent of the direction configured.	
0x09	**RegInterruptMask**	0xFF	7:0	Configures which [input-configured] IO will trigger an interrupt on NINT pin 0 : An event on this IO will trigger an interrupt 1 : An event on this IO will NOT trigger an interrupt	
0x0A	**RegSenseHigh**	0x00	7:6	Edge sensitivity of RegData[7]	00 : None 01 : Rising 10 : Falling 11 : Both
			5:4	Edge sensitivity of RegData[6]	
			3:2	Edge sensitivity of RegData[5]	
			1:0	Edge sensitivity of RegData[4]	
0x0B	**RegSenseLow**	0x00	7:6	Edge sensitivity of RegData[3]	00 : None 01 : Rising 10 : Falling 11 : Both
			5:4	Edge sensitivity of RegData[2]	
			3:2	Edge sensitivity of RegData[1]	
			1:0	Edge sensitivity of RegData[0]	
0x0C	**RegInterruptSource**	0x00	7:0	Interrupt source (from IOs set in RegInterruptMask) 0 : No interrupt has been triggered by this IO 1 : An interrupt has been triggered by this IO (an event as configured in relevant RegSense register occured). Writing '1' clears the bit in RegInterruptSource and in RegEventStatus When all bits are cleared, NINT signal goes back high.	
0x0D	**RegEventStatus**	0x00	7:0	Event status of all IOs. 0 : No event has occured on this IO 1 : An event has occured on this IO (an edge as configured in relevant RegSense register occured). Writing '1' clears the bit in RegEventStatus and in RegInterruptSource if relevant. If the edge sensitivity of the IO is changed, the bit(s) will be cleared automatically	
0x0E	**RegLevelShifter**	0x00	7:6	Level shifter mode for IO[3] (Bank A) and IO[7] (Bank B)	00 : OFF 01 : A->B 10 : B->A 11 : Reserved
			5:4	Level shifter mode for IO[2] (Bank A) and IO[6] (Bank B)	
			3:2	Level shifter mode for IO[1] (Bank A) and IO[5] (Bank B)	
			1:0	Level shifter mode for IO[0] (Bank A) and IO[4] (Bank B)	
0x0F	**RegClock**	0x00	7	Unused	
			6:5	Oscillator frequency (fOSC) source 00 : OFF. LED driver, keypad engine and debounce features are disabled. 01 : External clock input (OSCIN) 10 : Internal 2MHz oscillator 11 : Reserved	
			4	OSCIO pin function (Cf. §4.7) 0 : OSCIO is an input (OSCIN) 1 : OSCIO is an output (OSCOUT)	
			3:0	Frequency of the signal output on OSCOUT pin: 0x0 : 0Hz, permanent "0" logical level (GPO) 0xF : 0Hz, permanent "1" logical level (GPO) Else : fOSCOUT = fOSC/(2^(RegClock[3:0]-1))	
0x10	**RegMisc**	0x00	7	LED Driver mode for Bank B 's fading capable IOs (IO7) 0: Linear 1: Logarithmic	

			6:4	Frequency of the LED Driver clock ClkX of all IOs: 0 : OFF. LED driver functionality is disabled for all IOs. Else : ClkX = fOSC/(2^(RegMisc[6:4]-1))
			3	LED Driver mode for Bank A 's fading capable IOs (IO3) 0: Linear 1: Logarithmic
			2	NRESET pin function when externally forced low (Cf. §4.3.1 and §4.8.5). 0: Equivalent to POR 1: Reset PWM/Blink/Fade counters (not user programmed values) This bit is can only be reset manually or by POR, not by NRESET.
			1	Auto-increment register address (Cf. §4.4) 0: ON. When several consecutive data are read/written, register address is incremented. 1: OFF. When several consecutive data are read/written, register address is kept fixed.
			0	Autoclear NINT on RegData read (Cf. §4.6) 0: ON. RegInterruptSource is also automatically cleared when RegData is read. 1: OFF. RegInterruptSource must be manually cleared, either directly or via RegEventStatus.
0x11	**RegLEDDriverEnable**	0x00	7:0	Enables LED Driver for each [output-configured] IO 0 : LED Driver is disabled 1 : LED Driver is enabled
0x12	**RegDebounceConfig**	0x00	7:3	Unused
			2:0	Debounce time (Cf. §4.5.1) 000: 0.5ms x 2MHz/fOSC 001: 1ms x 2MHz/fOSC 010: 2ms x 2MHz/fOSC 011: 4ms x 2MHz/fOSC 100: 8ms x 2MHz/fOSC 101: 16ms x 2MHz/fOSC 110: 32ms x 2MHz/fOSC 111: 64ms x 2MHz/fOSC
0x13	**RegDebounceEnable**	0x00	7:0	Enables debouncing for each [input-configured] IO 0 : Debouncing is disabled 1 : Debouncing is enabled
0x14	**RegKeyConfig**	0x00	7	Unused
			6:5	Number of rows (outputs) + key scan enable 00 : Key scan OFF 01 : 2 rows – IO[0:1] 10 : 3 rows – IO[0:2] 11 : 4 rows – IO[0:3]
			4:3	Number of columns (inputs) 00 : 1 column – IO[4] 01 : 2 columns – IO[4:5] 10 : 3 columns – IO[4:6] 11 : 4 columns – IO[4:7]
			2:0	Scan time per row (must be set above debounce time). 000 : 1ms x 2MHz/fOSC 001 : 2ms x 2MHz/fOSC 010 : 4ms x 2MHz/fOSC 011 : 8ms x 2MHz/fOSC 100 : 16ms x 2MHz/fOSC 101 : 32ms x 2MHz/fOSC 110 : 64ms x 2MHz/fOSC 111 : 128ms x 2MHz/fOSC
0x15	**RegKeyData**	0xFF	7:0	Key which generated NINT (active low) Ex: RegKeyData=11011110 => key [IO5;IO0] has been pressed and generated NINT When read it is automatically cleared together with NINT and key scan continues.
0xXX	**RegTOnX**	0x00	7:5	Unused
			4:0	ON Time of IO[X]: 0 : Infinite (Static mode, TOn directly controlled by RegData, Cf §4.8.2) 1 - 15 : TOnX = 64 * RegTOnX * (255/ClkX) 16 - 31 : TOnX = 512 * RegTOnX * (255/ClkX)
0xXX	**RegIOnX**	0xFF	7:0	ON Intensity of IO[X] - Linear mode : IOnX = RegIOnX - Logarithmic mode (fading capable IOs only) : IOnX = f(RegIOnX) , Cf §4.8.5
0xXX	**RegOffX**	0x00	7:3	OFF Time of IO[X]: 0 : Infinite (Single shot mode, TOff directly controlled by RegData, Cf §4.8.3) 1 - 15 : TOffX = 64 * RegOffX[7:3] * (255/ClkX) 16 - 31 : TOffX = 512 * RegOffX[7:3] * (255/ClkX)
			2:0	OFF Intensity of IO[X] - Linear mode : IOffX = 4 x RegOff[2:0] - Logarithmic mode (fading capable IOs only) : IOffX = f(4 x RegOffX[2:0]) , Cf §4.8.5
0xXX	**RegTRiseX**	0x00	7:5	Unused
			4:0	Fade In setting of IO[X] 0 : OFF 1 - 15 : TRiseX = (RegIOnX-(4xRegOffX[2:0])) * RegTRiseX * (255/ClkX) 16 - 31 : TRiseX = 16 * (RegIOnX-(4xRegOffX[2:0])) * RegTRiseX * (255/ClkX)

0xXX	**RegTFallX**	0x00	7:5	Unused
			4:0	Fade Out setting of IO[X] 0 : OFF 1 - 15 : TFallX = (RegIOnX-(4xRegOffX[2:0])) * RegTFallX * (255/ClkX) 16 - 31 : TFallX = 16 * (RegIOnX-(4xRegOffX[2:0])) * RegTFallX * (255/ClkX)
0x2A	**RegHighInput**	0x00	7:0	Enables high input mode for each [input-configured] IO 0 : OFF. VIH max = 3.6V and VCCx min = 1.2V 1 : ON. VIH max = 5.5V and VCCx min = 1.65V
0x7D	**RegReset**	0x00	7:0	Software reset register Writing consecutively 0x12 and 0x34 will reset the device (same as POR). Always reads 0.

Table 9 – SX1508B Configuration Registers Description

5.2 SX1509B 16-channel GPIO with LED Driver and Keypad Engine

Address	Name	Description	Default
Device and IO Banks			
0x00	**RegInputDisableB**	Input buffer disable register - I/O[15-8] (Bank B)	0000 0000
0x01	**RegInputDisableA**	Input buffer disable register - I/O[7-0] (Bank A)	0000 0000
0x02	**RegLongSlewB**	Output buffer long slew register - I/O[15-8] (Bank B)	0000 0000
0x03	**RegLongSlewA**	Output buffer long slew register - I/O[7-0] (Bank A)	0000 0000
0x04	**RegLowDriveB**	Output buffer low drive register - I/O[15-8] (Bank B)	0000 0000
0x05	**RegLowDriveA**	Output buffer low drive register - I/O[7-0] (Bank A)	0000 0000
0x06	**RegPullUpB**	Pull-up register - I/O[15-8] (Bank B)	0000 0000
0x07	**RegPullUpA**	Pull-up register - I/O[7-0] (Bank A)	0000 0000
0x08	**RegPullDownB**	Pull-down register - I/O[15-8] (Bank B)	0000 0000
0x09	**RegPullDownA**	Pull-down register - I/O[7-0] (Bank A)	0000 0000
0x0A	**RegOpenDrainB**	Open drain register - I/O[15-8] (Bank B)	0000 0000
0x0B	**RegOpenDrainA**	Open drain register - I/O[7-0] (Bank A)	0000 0000
0x0C	**RegPolarityB**	Polarity register - I/O[15-8] (Bank B)	0000 0000
0x0D	**RegPolarityA**	Polarity register - I/O[7-0] (Bank A)	0000 0000
0x0E	**RegDirB**	Direction register - I/O[15-8] (Bank B)	1111 1111
0x0F	**RegDirA**	Direction register - I/O[7-0] (Bank A)	1111 1111
0x10	**RegDataB**	Data register - I/O[15-8] (Bank B)	1111 1111*
0x11	**RegDataA**	Data register - I/O[7-0] (Bank A)	1111 1111*
0x12	**RegInterruptMaskB**	Interrupt mask register - I/O[15-8] (Bank B)	1111 1111
0x13	**RegInterruptMaskA**	Interrupt mask register - I/O[7-0] (Bank A)	1111 1111
0x14	**RegSenseHighB**	Sense register for I/O[15:12]	0000 0000
0x15	**RegSenseLowB**	Sense register for I/O[11:8]	0000 0000
0x16	**RegSenseHighA**	Sense register for I/O[7:4]	0000 0000
0x17	**RegSenseLowA**	Sense register for I/O[3:0]	0000 0000
0x18	**RegInterruptSourceB**	Interrupt source register - I/O[15-8] (Bank B)	0000 0000
0x19	**RegInterruptSourceA**	Interrupt source register - I/O[7-0] (Bank A)	0000 0000
0x1A	**RegEventStatusB**	Event status register - I/O[15-8] (Bank B)	0000 0000
0x1B	**RegEventStatusA**	Event status register - I/O[7-0] (Bank A)	0000 0000
0x1C	**RegLevelShifter1**	Level shifter register	0000 0000
0x1D	**RegLevelShifter2**	Level shifter register	0000 0000
0x1E	**RegClock**	Clock management register	0000 0000
0x1F	**RegMisc**	Miscellaneous device settings register	0000 0000
0x20	**RegLEDDriverEnableB**	LED driver enable register - I/O[15-8] (Bank B)	0000 0000
0x21	**RegLEDDriverEnableA**	LED driver enable register - I/O[7-0] (Bank A)	0000 0000
Debounce and Keypad Engine			
0x22	**RegDebounceConfig**	Debounce configuration register	0000 0000
0x23	**RegDebounceEnableB**	Debounce enable register - I/O[15-8] (Bank B)	0000 0000
0x24	**RegDebounceEnableA**	Debounce enable register - I/O[7-0] (Bank A)	0000 0000
0x25	**RegKeyConfig1**	Key scan configuration register	0000 0000
0x26	**RegKeyConfig2**	Key scan configuration register	0000 0000
0x27	**RegKeyData1**	Key value (column)	1111 1111
0x28	**RegKeyData2**	Key value (row)	1111 1111
LED Driver (PWM, blinking, breathing)			
0x29	**RegTOn0**	ON time register for I/O[0]	0000 0000
0x2A	**RegIOn0**	ON intensity register for I/O[0]	1111 1111
0x2B	**RegOff0**	OFF time/intensity register for I/O[0]	0000 0000
0x2C	**RegTOn1**	ON time register for I/O[1]	0000 0000
0x2D	**RegIOn1**	ON intensity register for I/O[1]	1111 1111
0x2E	**RegOff1**	OFF time/intensity register for I/O[1]	0000 0000
0x2F	**RegTOn2**	ON time register for I/O[2]	0000 0000
0x30	**RegIOn2**	ON intensity register for I/O[2]	1111 1111
0x31	**RegOff2**	OFF time/intensity register for I/O[2]	0000 0000
0x32	**RegTOn3**	ON time register for I/O[3]	0000 0000
0x33	**RegIOn3**	ON intensity register for I/O[3]	1111 1111
0x34	**RegOff3**	OFF time/intensity register for I/O[3]	0000 0000
0x35	**RegTOn4**	ON time register for I/O[4]	0000 0000
0x36	**RegIOn4**	ON intensity register for I/O[4]	1111 1111
0x37	**RegOff4**	OFF time/intensity register for I/O[4]	0000 0000
0x38	**RegTRise4**	Fade in register for I/O[4]	0000 0000

Address	Name	Description	Default
0x39	**RegTFall4**	Fade out register for I/O[4]	0000 0000
0x3A	**RegTOn5**	ON time register for I/O[5]	0000 0000
0x3B	**RegIOn5**	ON intensity register for I/O[5]	1111 1111
0x3C	**RegOff5**	OFF time/intensity register for I/O[5]	0000 0000
0x3D	**RegTRise5**	Fade in register for I/O[5]	0000 0000
0x3E	**RegTFall5**	Fade out register for I/O[5]	0000 0000
0x3F	**RegTOn6**	ON time register for I/O[6]	0000 0000
0x40	**RegIOn6**	ON intensity register for I/O[6]	1111 1111
0x41	**RegOff6**	OFF time/intensity register for I/O[6]	0000 0000
0x42	**RegTRise6**	Fade in register for I/O[6]	0000 0000
0x43	**RegTFall6**	Fade out register for I/O[6]	0000 0000
0x44	**RegTOn7**	ON time register for I/O[7]	0000 0000
0x45	**RegIOn7**	ON intensity register for I/O[7]	1111 1111
0x46	**RegOff7**	OFF time/intensity register for I/O[7]	0000 0000
0x47	**RegTRise7**	Fade in register for I/O[7]	0000 0000
0x48	**RegTFall7**	Fade out register for I/O[7]	0000 0000
0x49	**RegTOn8**	ON time register for I/O[8]	0000 0000
0x4A	**RegIOn8**	ON intensity register for I/O[8]	1111 1111
0x4B	**RegOff8**	OFF time/intensity register for I/O[8]	0000 0000
0x4C	**RegTOn9**	ON time register for I/O[9]	0000 0000
0x4D	**RegIOn9**	ON intensity register for I/O[9]	1111 1111
0x4E	**RegOff9**	OFF time/intensity register for I/O[9]	0000 0000
0x4F	**RegTOn10**	ON time register for I/O[10]	0000 0000
0x50	**RegIOn10**	ON intensity register for I/O[10]	1111 1111
0x51	**RegOff10**	OFF time/intensity register for I/O[10]	0000 0000
0x52	**RegTOn11**	ON time register for I/O[11]	0000 0000
0x53	**RegIOn11**	ON intensity register for I/O[11]	1111 1111
0x54	**RegOff11**	OFF time/intensity register for I/O[11]	0000 0000
0x55	**RegTOn12**	ON time register for I/O[12]	0000 0000
0x56	**RegIOn12**	ON intensity register for I/O[12]	1111 1111
0x57	**RegOff12**	OFF time/intensity register for I/O[12]	0000 0000
0x58	**RegTRise12**	Fade in register for I/O[12]	0000 0000
0x59	**RegTFall12**	Fade out register for I/O[12]	0000 0000
0x5A	**RegTOn13**	ON time register for I/O[13]	0000 0000
0x5B	**RegIOn13**	ON intensity register for I/O[13]	1111 1111
0x5C	**RegOff13**	OFF time/intensity register for I/O[13]	0000 0000
0x5D	**RegTRise13**	Fade in register for I/O[13]	0000 0000
0x5E	**RegTFall13**	Fade out register for I/O[13]	0000 0000
0x5F	**RegTOn14**	ON time register for I/O[14]	0000 0000
0x60	**RegIOn14**	ON intensity register for I/O[14]	1111 1111
0x61	**RegOff14**	OFF time/intensity register for I/O[14]	0000 0000
0x62	**RegTRise14**	Fade in register for I/O[14]	0000 0000
0x63	**RegTFall14**	Fade out register for I/O[14]	0000 0000
0x64	**RegTOn15**	ON time register for I/O[15]	0000 0000
0x65	**RegIOn15**	ON intensity register for I/O[15]	1111 1111
0x66	**RegOff15**	OFF time/intensity register for I/O[15]	0000 0000
0x67	**RegTRise15**	Fade in register for I/O[15]	0000 0000
0x68	**RegTFall15**	Fade out register for I/O[15]	0000 0000
Miscellaneous			
0x69	**RegHighInputB**	High input enable register - I/O[15-8] (Bank B)	0000 0000
0x6A	**RegHighInputA**	High input enable register - I/O[7-0] (Bank A)	0000 0000
Software Reset			
0x7D	**RegReset**	Software reset register	0000 0000
Test (not to be written)			
0x7E	**RegTest1**	Test register	0000 0000
0x7F	**RegTest2**	Test register	0000 0000

*Bits set as output take "1" as default value.

Table 10 – SX1509B Configuration Registers Overview

Addr	Name	Default	Bits	Description
0x00	**RegInputDisableB**	0x00	7:0	Disables the input buffer of each IO 0 : Input buffer is enabled (input actually being used) 1 : Input buffer is disabled (input actually not being used or LED connection)
0x01	**RegInputDisableA**	0x00	7:0	Disables the input buffer of each IO 0 : Input buffer is enabled (input actually being used) 1 : Input buffer is disabled (input actually not being used, LED connection)
0x02	**RegLongSlewB**	0x00	7:0	Enables increased slew rate of the output buffer of each [output-configured] IO 0 : Increased slew rate is disabled 1 : Increased slew rate is enabled
0x03	**RegLongSlewA**	0x00	7:0	Enables increased slew rate of the output buffer of each [output-configured] IO 0 : Increased slew rate is disabled 1 : Increased slew rate is enabled
0x04	**RegLowDriveB**	0x00	7:0	Enables reduced drive of the output buffer of each [output-configured] IO 0 : Reduced drive is disabled 1 : Reduced drive is enabled. IOL specifications are divided by 2.
0x05	**RegLowDriveA**	0x00	7:0	Enables reduced drive of the output buffer of each [output-configured] IO 0 : Reduced drive is disabled 1 : Reduced drive is enabled. IOL specifications are divided by 2.
0x06	**RegPullUpB**	0x00	7:0	Enables the pull-up for each IO 0 : Pull-up is disabled 1 : Pull-up is enabled
0x07	**RegPullUpA**	0x00	7:0	Enables the pull-up for each IO 0 : Pull-up is disabled 1 : Pull-up is enabled
0x08	**RegPullDownB**	0x00	7:0	Enables the pull-down for each IO 0 : Pull-down is disabled 1 : Pull-down is enabled
0x09	**RegPullDownA**	0x00	7:0	Enables the pull-down for each IO 0 : Pull-down is disabled 1 : Pull-down is enabled
0x0A	**RegOpenDrainB**	0x00	7:0	Enables open drain operation for each [output-configured] IO 0 : Regular push-pull operation 1 : Open drain operation
0x0B	**RegOpenDrainA**	0x00	7:0	Enables open drain operation for each [output-configured] IO 0 : Regular push-pull operation 1 : Open drain operation
0x0C	**RegPolarityB**	0x00	7:0	Enables polarity inversion for each IO 0 : Normal polarity : RegData[x] = IO[x] 1 : Inverted polarity : RegData[x] = !IO[x] (for both input and output configured IOs)
0x0D	**RegPolarityA**	0x00	7:0	Enables polarity inversion for each IO 0 : Normal polarity : RegData[x] = IO[x] 1 : Inverted polarity : RegData[x] = !IO[x] (for both input and output configured IOs)
0x0E	**RegDirB**	0xFF	7:0	Configures direction for each IO. 0 : IO is configured as an output 1 : IO is configured as an input
0x0F	**RegDirA**	0xFF	7:0	Configures direction for each IO. 0 : IO is configured as an output 1 : IO is configured as an input
0x10	**RegDataB**	0xFF	7:0	Write: Data to be output to the output-configured IOs Read: Data seen at the IOs, independent of the direction configured.
0x11	**RegDataA**	0xFF	7:0	Write: Data to be output to the output-configured IOs Read: Data seen at the IOs, independent of the direction configured.
0x12	**RegInterruptMaskB**	0xFF	7:0	Configures which [input-configured] IO will trigger an interrupt on NINT pin 0 : An event on this IO will trigger an interrupt 1 : An event on this IO will NOT trigger an interrupt
0x13	**RegInterruptMaskA**	0xFF	7:0	Configures which [input-configured] IO will trigger an interrupt on NINT pin 0 : An event on this IO will trigger an interrupt 1 : An event on this IO will NOT trigger an interrupt
0x14	**RegSenseHighB**	0x00	7:6	Edge sensitivity of RegData[15]	00 : None 01 : Rising 10 : Falling 11 : Both
			5:4	Edge sensitivity of RegData[14]	
			3:2	Edge sensitivity of RegData[13]	
			1:0	Edge sensitivity of RegData[12]	
0x15	**RegSenseLowB**	0x00	7:6	Edge sensitivity of RegData[11]	00 : None 01 : Rising 10 : Falling 11 : Both
			5:4	Edge sensitivity of RegData[10]	
			3:2	Edge sensitivity of RegData[9]	
			1:0	Edge sensitivity of RegData[8]	
0x16	**RegSenseHighA**	0x00	7:6	Edge sensitivity of RegData[7]	00 : None 01 : Rising 10 : Falling 11 : Both
			5:4	Edge sensitivity of RegData[6]	
			3:2	Edge sensitivity of RegData[5]	
			1:0	Edge sensitivity of RegData[4]	
0x17	**RegSenseLowA**	0x00	7:6	Edge sensitivity of RegData[3]	00 : None

			5:4	Edge sensitivity of RegData[2]	
			3:2	Edge sensitivity of RegData[1]	
			1:0	Edge sensitivity of RegData[0]	
0x18	**RegInterruptSourceB**	0x00	7:0	Interrupt source (from IOs set in RegInterruptMask) 0 : No interrupt has been triggered by this IO 1 : An interrupt has been triggered by this IO (an event as configured in relevant RegSense register occured). Writing '1' clears the bit in RegInterruptSource and in RegEventStatus When all bits are cleared, NINT signal goes back high.	
0x19	**RegInterruptSourceA**	0x00	7:0	Interrupt source (from IOs set in RegInterruptMask) 0 : No interrupt has been triggered by this IO 1 : An interrupt has been triggered by this IO (an event as configured in relevant RegSense register occured). Writing '1' clears the bit in RegInterruptSource and in RegEventStatus When all bits are cleared, NINT signal goes back high.	
0x1A	**RegEventStatusB**	0x00	7:0	Event status of all IOs. 0 : No event has occured on this IO 1 : An event has occured on this IO (an edge as configured in relevant RegSense register occured). Writing '1' clears the bit in RegEventStatus and in RegInterruptSource if relevant. If the edge sensitivity of the IO is changed, the bit(s) will be cleared automatically	
0x1B	**RegEventStatusA**	0x00	7:0	Event status of all IOs. 0 : No event has occured on this IO 1 : An event has occured on this IO (an edge as configured in relevant RegSense register occured). Writing '1' clears the bit in RegEventStatus and in RegInterruptSource if relevant. If the edge sensitivity of the IO is changed, the bit(s) will be cleared automatically	
0x1C	**RegLevelShifter1**	0x00	7:6	Level shifter mode for IO[7] (Bank A) and IO[15] (Bank B)	00 : OFF 01 : A->B 10 : B->A 11 : Reserved
			5:4	Level shifter mode for IO[6] (Bank A) and IO[14] (Bank B)	
			3:2	Level shifter mode for IO[5] (Bank A) and IO[13] (Bank B)	
			1:0	Level shifter mode for IO[4] (Bank A) and IO[12] (Bank B)	
0x1D	**RegLevelShifter2**	0x00	7:6	Level shifter mode for IO[3] (Bank A) and IO[11] (Bank B)	00 : OFF 01 : A->B 10 : B->A 11 : Reserved
			5:4	Level shifter mode for IO[2] (Bank A) and IO[10] (Bank B)	
			3:2	Level shifter mode for IO[1] (Bank A) and IO[9] (Bank B)	
			1:0	Level shifter mode for IO[0] (Bank A) and IO[8] (Bank B)	
0x1E	**RegClock**	0x00	7	Unused	
			6:5	Oscillator frequency (fOSC) source 00 : OFF. LED driver, keypad engine and debounce features are disabled. 01 : External clock input (OSCIN) 10 : Internal 2MHz oscillator 11 : Reserved	
			4	OSCIO pin function (Cf. §4.7) 0 : OSCIO is an input (OSCIN) 1 : OSCIO is an output (OSCOUT)	
			3:0	Frequency of the signal output on OSCOUT pin: 0x0 : 0Hz, permanent "0" logical level (GPO) 0xF : 0Hz, permanent "1" logical level (GPO) Else : fOSCOUT = fOSC/(2^(RegClock[3:0]-1))	
0x1F	**RegMisc**	0x00	7	LED Driver mode for Bank B 's fading capable IOs (IO15-12) 0: Linear 1: Logarithmic	
			6:4	Frequency of the LED Driver clock ClkX of all IOs: 0 : OFF. LED driver functionality is disabled for all IOs. Else : ClkX = fOSC/(2^(RegMisc[6:4]-1))	
			3	LED Driver mode for Bank A 's fading capable IOs (IO7-4) 0: Linear 1: Logarithmic	
			2	NRESET pin function when externally forced low (Cf. §4.3.1 and §4.8.5) 0: Equivalent to POR 1: Reset PWM/Blink/Fade counters (not user programmed values) This bit is can only be reset manually or by POR, not by NRESET.	
			1	Auto-increment register address (Cf. §4.4) 0: ON. When several consecutive data are read/written, register address is incremented. 1: OFF. When several consecutive data are read/written, register address is kept fixed.	
			0	Autoclear NINT on RegData read (Cf. §4.6) 0: ON. RegInterruptSourceA/B is also automatically cleared when RegDataA/B is read. 1: OFF. RegInterruptSourceA/B must be manually cleared, either directly or via RegEventStatusA/B.	
0x20	**RegLEDDriverEnableB**	0x00	7:0	Enables LED Driver for each [output-configured] IO 0 : LED Driver is disabled 1 : LED Driver is enabled	

0x21	**RegLEDDriverEnableA**	0x00	7:0	Enables LED Driver for each [output-configured] IO 0 : LED Driver is disabled 1 : LED Driver is enabled
0x22	**RegDebounceConfig**	0x00	7:3	Unused
			2:0	Debounce time (Cf. §4.5.1) 000: 0.5ms x 2MHz/fOSC 001: 1ms x 2MHz/fOSC 010: 2ms x 2MHz/fOSC 011: 4ms x 2MHz/fOSC 100: 8ms x 2MHz/fOSC 101: 16ms x 2MHz/fOSC 110: 32ms x 2MHz/fOSC 111: 64ms x 2MHz/fOSC
0x23	**RegDebounceEnableB**	0x00	7:0	Enables debouncing for each [input-configured] IO 0 : Debouncing is disabled 1 : Debouncing is enabled
0x24	**RegDebounceEnableA**	0x00	7:0	Enables debouncing for each [input-configured] IO 0 : Debouncing is disabled 1 : Debouncing is enabled
0x25	**RegKeyConfig1**	0x00	7	Reserved
			6:4	Auto Sleep time (no key press within this time will set keypad engine to sleep) 000 : OFF 001 : 128ms x 2MHz/fOSC 010 : 256ms x 2MHz/fOSC 011 : 512ms x 2MHz/fOSC 100 : 1sec x 2MHz/fOSC 101 : 2sec x 2MHz/fOSC 110 : 4sec x 2MHz/fOSC 111 : 8sec x 2MHz/fOSC
			3	Unused
			2:0	Scan time per row (must be set above debounce time). 000 : 1ms x 2MHz/fOSC 001 : 2ms x 2MHz/fOSC 010 : 4ms x 2MHz/fOSC 011 : 8ms x 2MHz/fOSC 100 : 16ms x 2MHz/fOSC 101 : 32ms x 2MHz/fOSC 110 : 64ms x 2MHz/fOSC 111 : 128ms x 2MHz/fOSC
0x26	**RegKeyConfig2**	0x00	7:6	Unused
			5:3	Number of rows (outputs) + key scan enable 000 : Key scan OFF 001 : 2 rows – IO[0:1] 010 : 3 rows – IO[0:2] 011 : 4 rows – IO[0:3] 100 : 5 rows – IO[0:4] 101 : 6 rows – IO[0:5] 110 : 7 rows – IO[0:6] 111 : 8 rows – IO[0:7]
			2:0	Number of columns (inputs) 000 : 1 column – IO[8] 001 : 2 columns – IO[8:9] 010 : 3 columns – IO[8:10] 011 : 4 columns – IO[8:11] 100 : 5 columns – IO[8:12] 101 : 6 columns – IO[8:13] 110 : 7 columns – IO[8:14] 111 : 8 columns – IO[8:15]
0x27	**RegKeyData1**	0xFF	7:0	Column which generated NINT (active low) Ex: RegKeyData1=11011111 => IO13 has generated NINT The register is automatically cleared when RegKeyData2 is read.
0x28	**RegKeyData2**	0xFF	7:0	Row which generated NINT (active low) Ex: RegKeyData2=11111110 => IO0 has generated NINT When the register is read both RegKeyData1 & RegKeyData2 are automatically cleared together with NINT and key scan continues.
0xXX	**RegTOnX**	0x00	7:5	Unused
			4:0	ON Time of IO[X]: 0 : Infinite (Static mode, TOn directly controlled by RegData, Cf §4.8.2) 1 - 15 : TOnX = 64 * RegTOnX * (255/ClkX) 16 - 31 : TOnX = 512 * RegTOnX * (255/ClkX)
0xXX	**RegIOnX**	0xFF	7:0	ON Intensity of IO[X] - Linear mode : IOnX = RegIOnX - Logarithmic mode (fading capable IOs only) : IOnX = f(RegIOnX) , Cf §4.8.5

0xXX	**RegOffX**	0x00	7:3	OFF Time of IO[X]: 0 : Infinite (Single shot mode, TOff directly controlled by RegData, Cf §4.8.3) 1 - 15 : TOffX = 64 * RegOffX[7:3] * (255/ClkX) 16 - 31 : TOffX = 512 * RegOffX[7:3] * (255/ClkX)
			2:0	OFF Intensity of IO[X] - Linear mode : IOffX = 4 x RegOff[2:0] - Logarithmic mode (fading capable IOs only) : IOffX = f(4 x RegOffX[2:0]) , Cf §4.8.5
0xXX	**RegTRiseX**	0x00	7:5	Unused
			4:0	Fade In setting of IO[X] 0 : OFF 1 - 15 : TRiseX = (RegIOnX-(4xRegOffX[2:0])) * RegTRiseX * (255/ClkX) 16 - 31 : TRiseX = 16 * (RegIOnX-(4xRegOffX[2:0])) * RegTRiseX * (255/ClkX)
0xXX	**RegTFallX**	0x00	7:5	Unused
			4:0	Fade Out setting of IO[X] 0 : OFF 1 - 15 : TFallX = (RegIOnX-(4xRegOffX[2:0])) * RegTFallX * (255/ClkX) 16 - 31 : TFallX = 16 * (RegIOnX-(4xRegOffX[2:0])) * RegTFallX * (255/ClkX)
0x69	**RegHighInputB**	0x00	7:0	Enables high input mode for each [input-configured] IO 0 : OFF. VIH max = 3.6V and VCCx min = 1.2V 1 : ON. VIH max = 5.5V and VCCx min = 1.65V
0x6A	**RegHighInputA**	0x00	7:0	Enables high input mode for each [input-configured] IO 0 : OFF. VIH max = 3.6V and VCCx min = 1.2V 1 : ON. VIH max = 5.5V and VCCx min = 1.65V
0x7D	**RegReset**	0x00	7:0	Software reset register Writing consecutively 0x12 and 0x34 will reset the device (same as POR). Always reads 0.

Table 11 – SX1509B Configuration Registers Description

6 APPLICATION INFORMATION

6.1 Typical Application Circuit

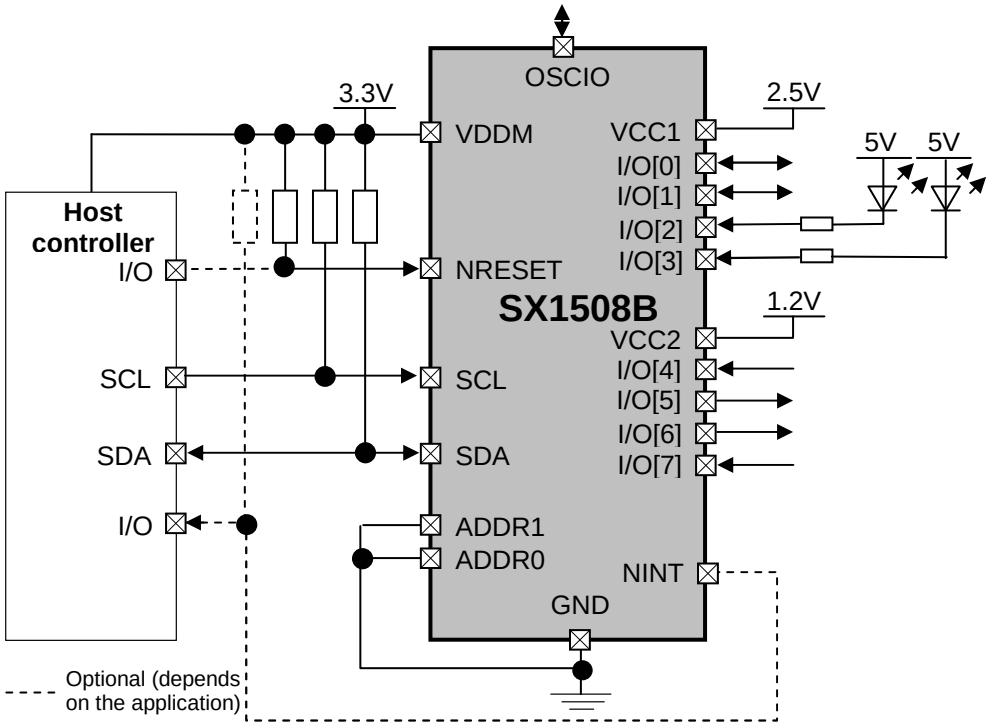


Figure 19 - Typical Application Schematic

6.2 Typical LED Connection

Typical LED Connection is described below. The LED is usually connected to a high voltage (VBAT) to take advantage of the high sink current of the I/O and to accommodate high LED threshold voltages (VLED).

Please note that in this configuration the IO must be programmed as open drain output (RegOpenDrain) with no pull-up (RegPullUp) and input buffer must be disabled (RegInputBufferDisable).

VCCx can take any value without compromising LED operation.

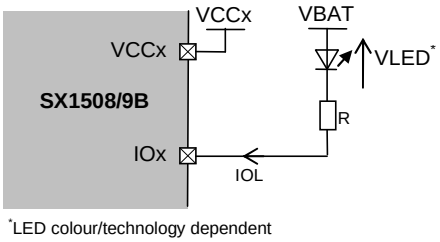


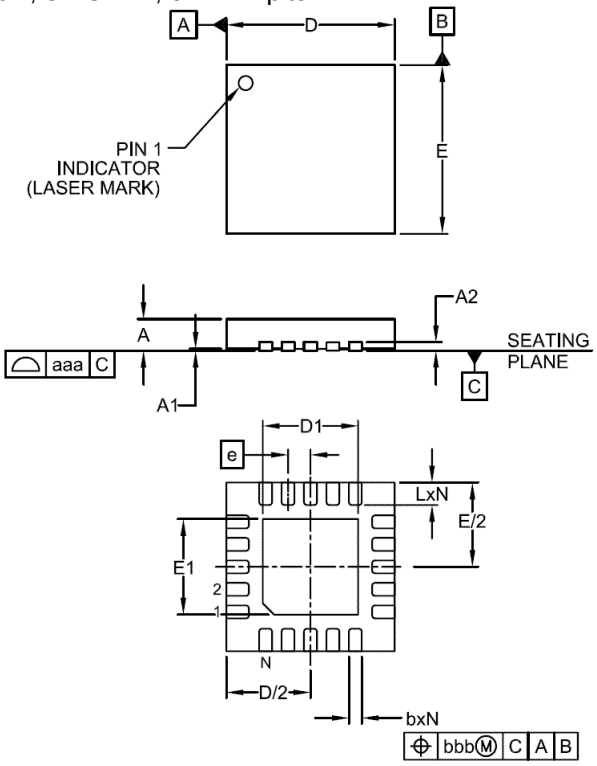
Figure 20 – Typical LED Operation

Serial R must be calculated for IOL not to exceed its max spec (Cf. Table 5) else VOL will increase.

7 PACKAGING INFORMATION

7.1 QFN-UT 20-pin Outline Drawing

QFN-UT 20-pin, 3 x 3 mm, 0.4 mm pitch



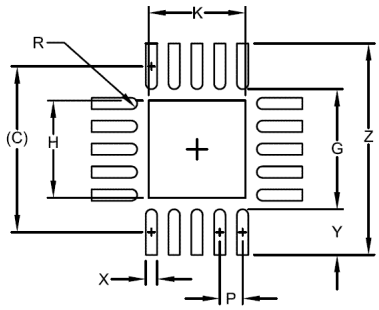
DIM	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX
A	.020	-	.024	0.50	-	0.60
A1	.000	-	.002	0.00	-	0.05
A2		(.006)			(0.152)	
b	.006	.008	.010	0.15	0.20	0.25
D	.114	.118	.122	2.90	3.00	3.10
D1	.061	.067	.071	1.55	1.70	1.80
E	.114	.118	.122	2.90	3.00	3.10
E1	.061	.067	.071	1.55	1.70	1.80
e		.016 BSC			0.40 BSC	
L	.012	.016	.020	0.30	0.40	0.50
N		20			20	
aaa		.003			0.08	
bbb		.004			0.10	

NOTES:

1. CONTROLLING DIMENSIONS ARE IN MILLIMETERS (ANGLES IN DEGREES).
2. COPLANARITY APPLIES TO THE EXPOSED PAD AS WELL AS THE TERMINALS.
3. DAP IS 1.90 x 1.90mm.

Figure 21 - QFN-UT 20-pin Outline Drawing

7.2 QFN-UT 20-pin Land Pattern



DIM	INCHES	MILLIMETERS
C	(.114)	(2.90)
G	.083	2.10
H	.067	1.70
K	.067	1.70
P	.016	0.40
R	.004	0.10
X	.008	0.20
Y	.031	0.80
Z	.146	3.70

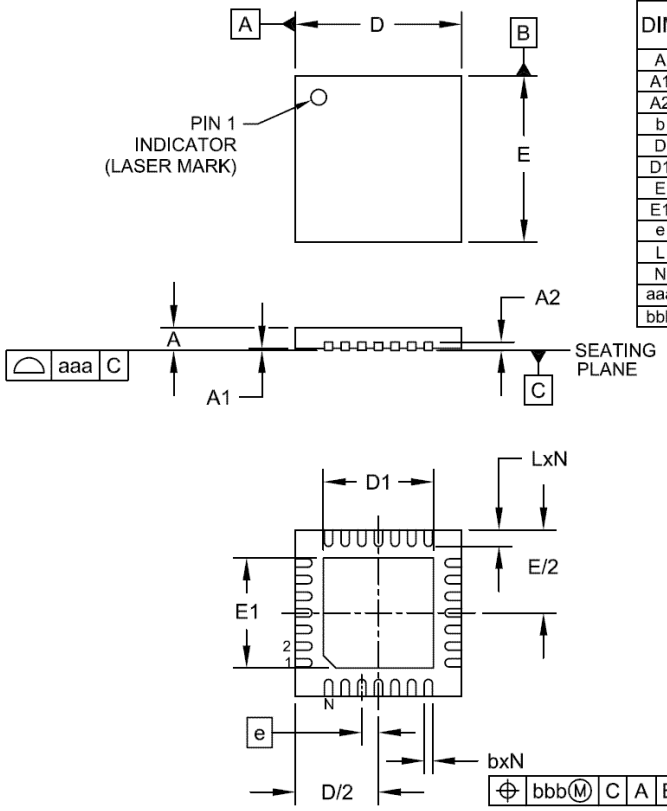
NOTES:

1. CONTROLLING DIMENSIONS ARE IN MILLIMETERS (ANGLES IN DEGREES).
2. THIS LAND PATTERN IS FOR REFERENCE PURPOSES ONLY. CONSULT YOUR MANUFACTURING GROUP TO ENSURE YOUR COMPANY'S MANUFACTURING GUIDELINES ARE MET.
3. THERMAL VIAS IN THE LAND PATTERN OF THE EXPOSED PAD SHALL BE CONNECTED TO A SYSTEM GROUND PLANE. FAILURE TO DO SO MAY COMPROMISE THE THERMAL AND/OR FUNCTIONAL PERFORMANCE OF THE DEVICE.

Figure 22 - QFN-UT 20-pin Land Pattern

7.3 QFN-UT 28-pin Outline Drawing

QFN-UT 28-pin, 4 x 4 mm, 0.4 mm pitch



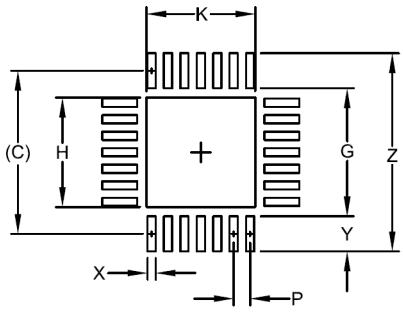
DIM	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX
A	.020	-	.024	0.50	-	0.60
A1	.000	-	.001	0.00	-	0.02
A2		(.006)			(0.152)	
b	.006	.008	.010	0.15	0.20	0.25
D	.154	.157	.161	3.90	4.00	4.10
D1	.100	.104	.108	2.55	2.65	2.75
E	.154	.157	.161	3.90	4.00	4.10
E1	.100	.104	.108	2.55	2.65	2.75
e		.016 BSC			0.40 BSC	
L	.012	.016	.020	0.30	0.40	0.50
N		28			28	
aaa		.003			0.08	
bbb		.004			0.10	

NOTES:

1. CONTROLLING DIMENSIONS ARE IN MILLIMETERS (ANGLES IN DEGREES).
2. COPLANARITY APPLIES TO THE EXPOSED PAD AS WELL AS THE TERMINALS.

Figure 23 - QFN-UT 28-pin Outline Drawing

7.4 QFN-UT 28-pin Land Pattern



DIM	INCHES	MILLIMETERS
C	(.156)	(3.95)
G	.122	3.10
H	.104	2.65
K	.104	2.65
P	.016	0.40
X	.008	0.20
Y	.033	0.85
Z	.189	4.80

NOTES:

1. CONTROLLING DIMENSIONS ARE IN MILLIMETERS (ANGLES IN DEGREES).
2. THIS LAND PATTERN IS FOR REFERENCE PURPOSES ONLY. CONSULT YOUR MANUFACTURING GROUP TO ENSURE YOUR COMPANY'S MANUFACTURING GUIDELINES ARE MET.
3. THERMAL VIAS IN THE LAND PATTERN OF THE EXPOSED PAD SHALL BE CONNECTED TO A SYSTEM GROUND PLANE. FAILURE TO DO SO MAY COMPROMISE THE THERMAL AND/OR FUNCTIONAL PERFORMANCE OF THE DEVICE.
4. SQUARE PACKAGE-DIMENSIONS APPLY IN BOTH X AND Y DIRECTIONS.

Figure 24 - QFN-UT 28-pin Land Pattern

8 SOLDERING PROFILE

The soldering reflow profile for the SX1508B and SX1509B is described in the standard IPC/JEDEC J-STD-020C. For detailed information please go to http://www.jedec.org/download/search/jstd020c.pdf

Profile Feature	Sn-Pb Eutectic Assembly	Pb-Free Assembly
Average Ramp-Up Rate (Ts_{max} to Tp)	3 °C/second max.	3° C/second max.
Preheat – Temperature Min (Ts_{min}) – Temperature Max (Ts_{max}) – Time (ts_{min} to ts_{max})	 100 °C 150 °C 60-120 seconds	 150 °C 200 °C 60-180 seconds
Time maintained above: – Temperature (T_L) – Time (t_L)	 183 °C 60-150 seconds	 217 °C 60-150 seconds
Peak/Classification Temperature (Tp)	See Table 4.1	See Table 4.2
Time within 5 °C of actual Peak Temperature (tp)	10-30 seconds	20-40 seconds
Ramp-Down Rate	6 °C/second max.	6 °C/second max.
Time 25 °C to Peak Temperature	6 minutes max.	8 minutes max.

Note 1: All temperatures refer to topside of the package, measured on the package body surface.

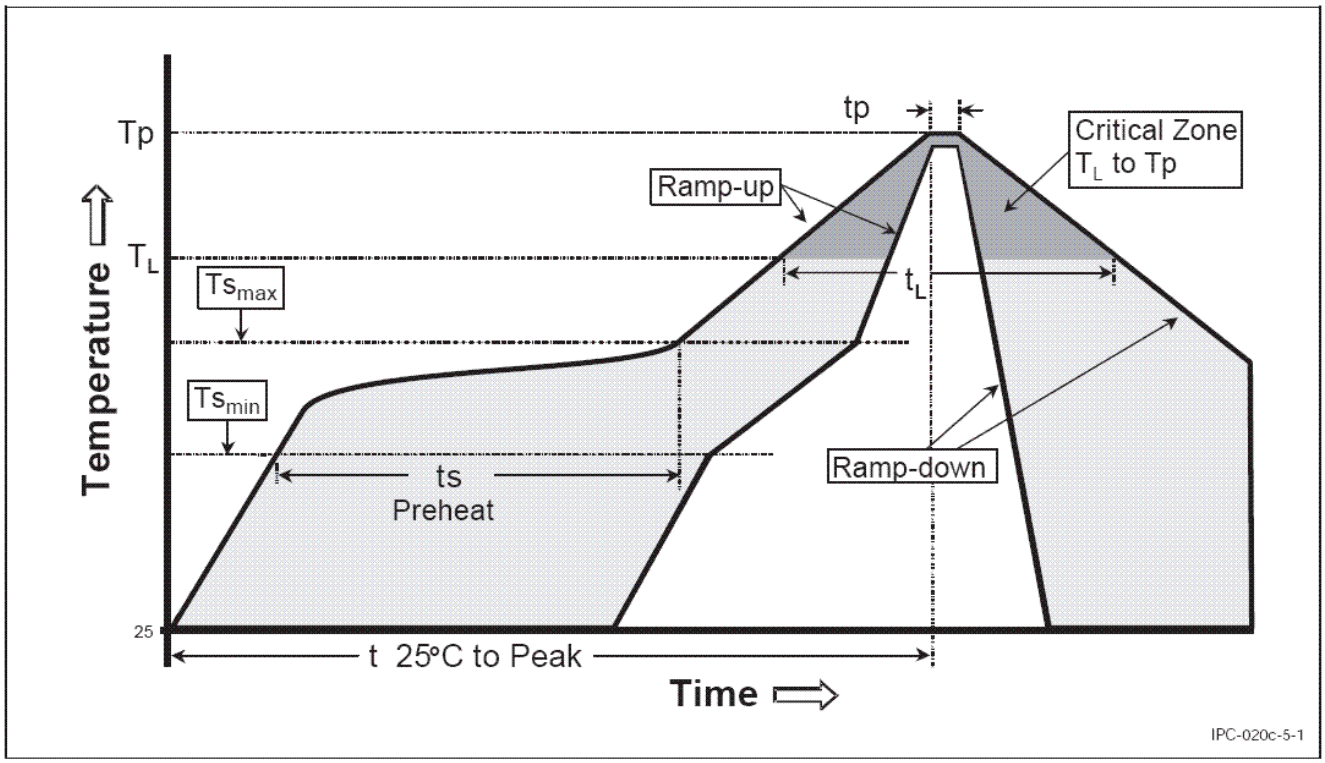


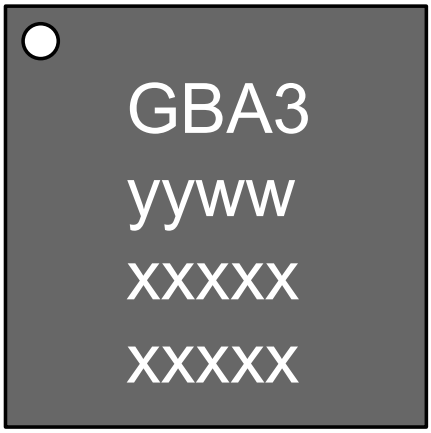
Figure 25 - Classification Reflow Profile (IPC/JEDEC J-STD-020C)

9 MARKING INFORMATION



yyww = Date Code
xxxx = Semtech Lot No.

Figure 26 – SX1508B Marking Information



yyww = Date Code
xxxxx = Semtech Lot No.
xxxxx

Figure 27 – SX1509B Marking Information

© Semtech 2011
All rights reserved. Reproduction in whole or in part is prohibited without the prior written consent of the copyright owner. The information presented in this document does not form part of any quotation or contract, is believed to be accurate and reliable and may be changed without notice. No liability will be accepted by the publisher for any consequence of its use. Publication thereof does not convey nor imply any license under patent or other industrial or intellectual property rights. Semtech assumes no responsibility or liability whatsoever for any failure or unexpected operation resulting from misuse, neglect improper installation, repair or improper handling or unusual physical or electrical stress including, but not limited to, exposure to parameters beyond the specified maximum ratings or operation outside the specified range.

SEMTECH PRODUCTS ARE NOT DESIGNED, INTENDED, AUTHORIZED OR WARRANTED TO BE SUITABLE FOR USE IN LIFE-SUPPORT APPLICATIONS, DEVICES OR SYSTEMS OR OTHER CRITICAL APPLICATIONS. INCLUSION OF SEMTECH PRODUCTS IN SUCH APPLICATIONS IS UNDERSTOOD TO BE UNDERTAKEN SOLELY AT THE CUSTOMER'S OWN RISK. Should a customer purchase or use Semtech products for any such unauthorized application, the customer shall indemnify and hold Semtech and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs damages and attorney fees which could arise.

Notice: All referenced brands, product names, service names and trademarks are the property of their respective owners.

Contact Information

Semtech Corporation
Advanced Communications and Sensing Products Division
200 Flynn Road, Camarillo, CA 93012
Phone: (805) 498-2111 Fax: (805) 498-3804